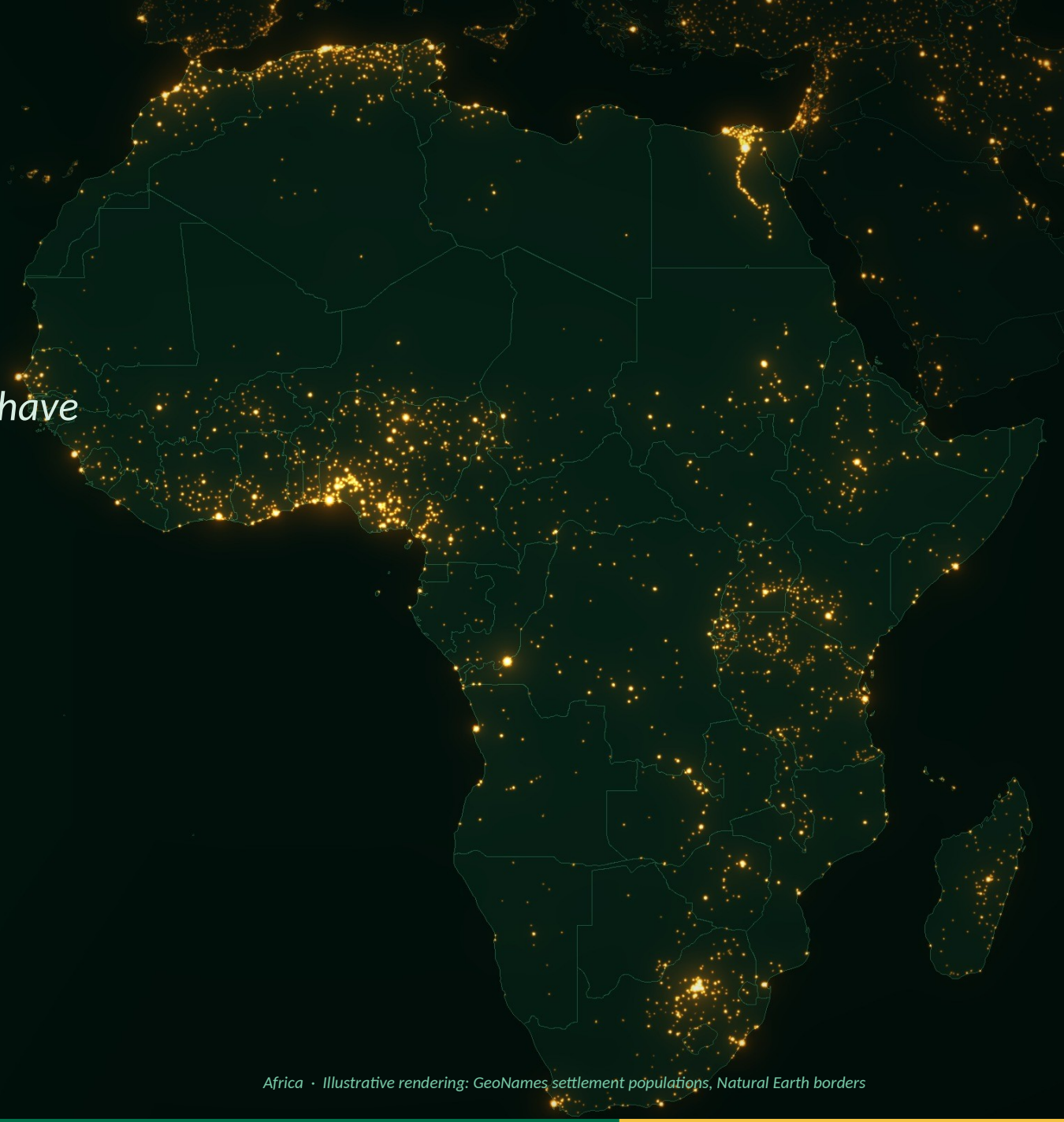
AFRICAN DEVELOPMENT BANK · DATA INNOVATION FOR AFRICA (DI4A)

Night-Time Lights

Measuring activity from orbit — and what lights have been used to measure

Night-time lights for macroeconomic statistics





Africa · Illustrative rendering: GeoNames settlement populations, Natural Earth borders

Two sessions and what you will be able to do

SESSION A · 45 MIN

Night-time lights: measuring activity from orbit

- Collect NASA Black Marble data and clip a monthly composite for a selected country
- Identify what night-time lights proxy well — and what they do not
- Review key artefacts: blooming, saturation, gas flares, seasonality, cloud cover, sensor discontinuity
- Apply masks, aggregate over time and compare regions
- Produce a national series with notes on data limitations

SESSION B · 90 MIN

Night-time lights in practice: what they have been used to measure

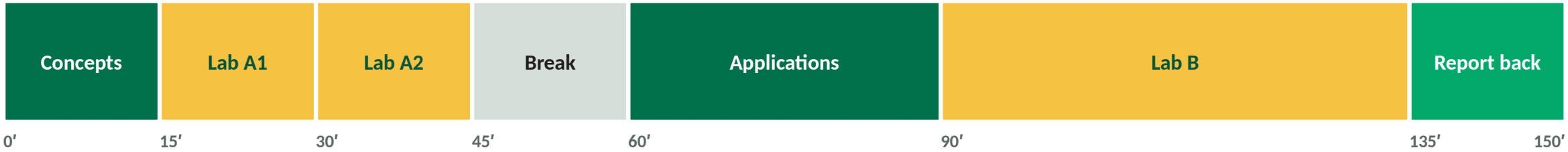
- Review common applications: output proxies, electrification, urbanisation, damage assessment, activity mapping
- Assess how each use case was validated and where the proxy performs well
- Rebuild one application and compare it with an official indicator
- Report where the relationship holds — and where it breaks

By the end you will hold three outputs: ① a clipped monthly composite for your country ② a masked national series with a limitations note ③ one rebuilt application, compared with an official indicator.

How the sessions run: concepts, then hands-on, twice

SESSION A

SESSION B



Concepts · 15 min

Fast tour of the physics, sensors and products. Reference slides are flagged — return to them during the labs.

Lab A · 30 min

Download, clip, mask and aggregate one country. Output: a monthly national series plus a short limitations note.

Lab B · 45 min

Choose output, electrification, urbanisation or shock. Compare your NTL indicator with an official series.

Report back · 15 min

Each table states in one sentence where the relationship held, where it broke, and why.

Afterwards The boundaries, grids and zonal statistics produced here can be reused for cropland and built-up analyses.

SECTION 1

01

Foundations

Why should a photograph of the dark tell us anything about an economy?

The core idea · Definitions · The inferential chain · What the sensor sees · Why it matters

SESSION A · ≈ 5 MIN

Maghreb coast · Illustrative rendering: GeoNames settlement populations, Natural Earth borders

Light is what human activity leaks

Almost everything people do after sunset requires light — and part of that light escapes upward, into the field of view of a satellite.

That physical fact turns the night side of the planet into an observation of human presence that no census, survey or administrative register can match on all four properties at once:

Globally consistent

Same method in every country

Spatially continuous

Every ≈ 500 m grid cell

Politically neutral

Independent of reporting

Freely available

Open NASA / EOG products

1992

Start of the continuous digital DMSP record — three decades of history

≈ 500 m

Black Marble grid (15 arc-seconds); native VIIRS DNB pixel ≈ 742 m, near-constant across the swath

Nightly

Global VIIRS overpass around 01:30 local time, since 2012

For a statistical office: censuses come roughly once a decade, surveys every few years, quarterly GDP with a lag — lights arrive every night, for every district.

What night-time lights are — and are not

WORKING DEFINITION

Night-time lights (NTL) are measurements of visible and near-infrared radiance travelling upward from the Earth's surface, recorded by a satellite sensor during local night, when the scene is free of direct sunlight.

What it is

A physical measurement

Radiance in $nW \cdot cm^{-2} \cdot sr^{-1}$ for calibrated sensors (VIIRS).

A consistent grid

Same method over every country — small islands and large states alike.

A near-real-time record

Daily since 2012; monthly and annual composites, free of charge.

What it is not

A photograph

The familiar "Earth at night" is a statistical composite of many cloud-free nights.

A direct measure of people or income

Those are inferences — each needs its own justification.

A complete record of lighting

Indoor, shielded and sideways light never reaches the sensor.

From electricity to a pixel value: five links

1

Energy & infrastructure

A grid, generator or fuel supply exists and is operating.

2

Activity after dark

People use that energy for lighting: streets, homes, markets, industry.

3

Upward emission

A fraction of the light escapes upward rather than being shielded.

4

Atmospheric transfer

Photons cross the atmosphere, attenuated by aerosols, vapour and cloud.

5

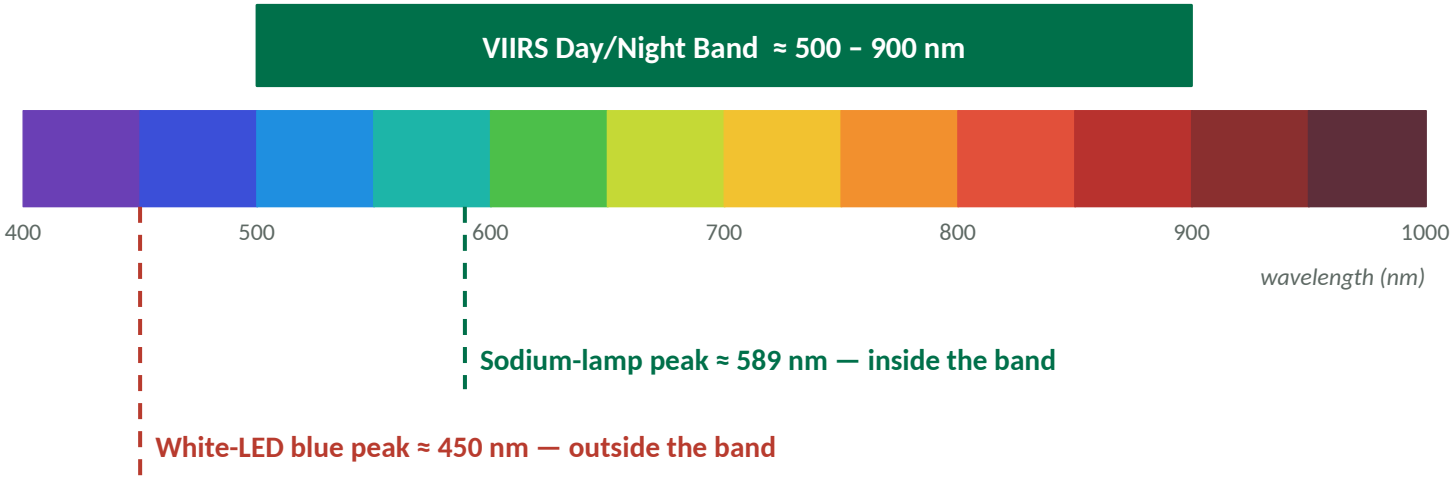
Sensor response

The detector integrates them over its band into a radiance value.

Where the chain breaks in African contexts — a dark pixel can mean …

- **Link 1** — Load-shedding or fuel shortage — people are there, power is not
- **Link 2–3** — Solar lanterns and small home systems too faint to detect
- **Link 4** — Rainy-season cloud, harmattan dust, biomass-burning smoke
- **Link 5** — Rural light below the sensor's detection floor

The sensor sees one band, at one hour of the night



Why this matters for statistics

Two cities with the same economic activity can produce different radiance because of lamp type, curfew habits or building form. Changes in lighting technology — Africa's accelerating LED and solar roll-out — can move the series without any change in activity. Treat technology change as a known source of drift and prefer within-unit changes over short windows.

The LED paradox

Swapping sodium for white LED street lights can make a city look dimmer to VIIRS while it is actually better lit on the ground.

The 01:30 snapshot

VIIRS passes after midnight. Evening markets, traffic and commerce (18:00–22:00) are largely unobserved; street lighting dominates.

Only upward light

Shielded fixtures, indoor lighting and light blocked by trees or buildings never reach the sensor.

What lights add to macroeconomic statistics

What lights add

Timeliness

Monthly composites are typically available within weeks; daily products within days — ahead of quarterly national accounts.

Granularity

A 500 m grid can be aggregated to any boundary: province, district, city, corridor or economic zone.

Coverage of the unrecorded

Informal and unregistered activity still needs light; it shows up even where registers and surveys miss it.

Independent cross-check

A source no national institution produces — useful to benchmark, flag anomalies and support credibility.

What lights cannot give you

- **Value added by sector.** Agriculture, mining and many services create value with little light.
- **Prices or deflators.** Lights are a volume-like signal at best.
- **Household income.** Bright pixels are not people, and not their incomes.
- **Daytime activity.** Markets, farms and offices that close before dark are invisible.

Positioning: a complement to official statistics for disaggregation, nowcasting and verification — never a replacement for national accounts, surveys or the census.

SECTION 2

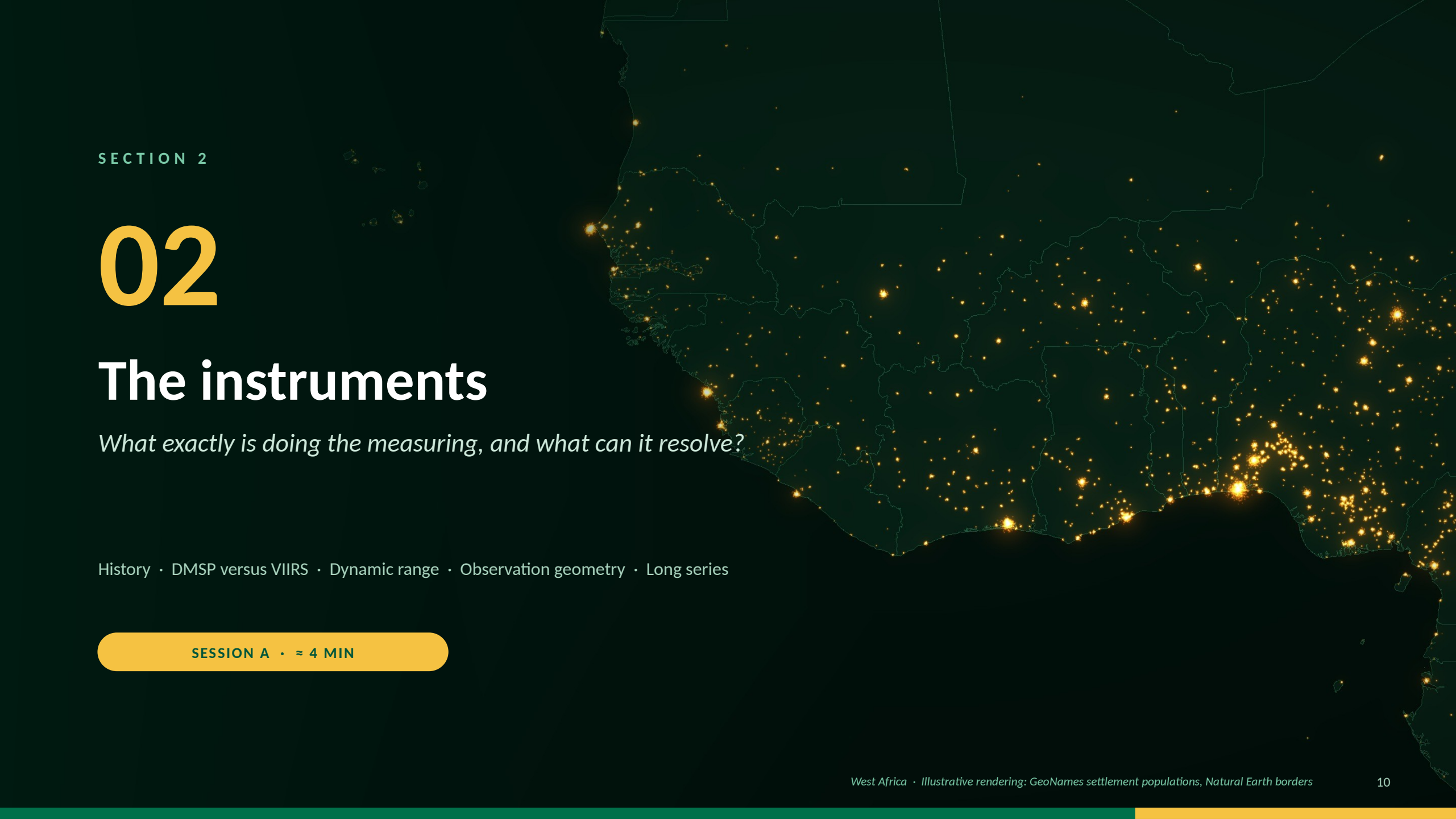
02

The instruments

What exactly is doing the measuring, and what can it resolve?

History · DMSP versus VIIRS · Dynamic range · Observation geometry · Long series

SESSION A · ≈ 4 MIN



West Africa · Illustrative rendering: GeoNames settlement populations, Natural Earth borders

Fifty years of looking down at the dark

1970s — DMSP-OLS night imagery (film). City lights noticed as a by-product of cloud detection.

1992 — Start of the digital DMSP archive → "stable lights" annual series to 2013.

2011 — Suomi NPP launched with VIIRS: first sensor calibrated for night-time radiance.

2012 — VIIRS monthly cloud-free composites; NASA Black Marble record begins (Jan 2012).

2017 — NOAA-20 launched with a second VIIRS; Black Marble twin products follow.

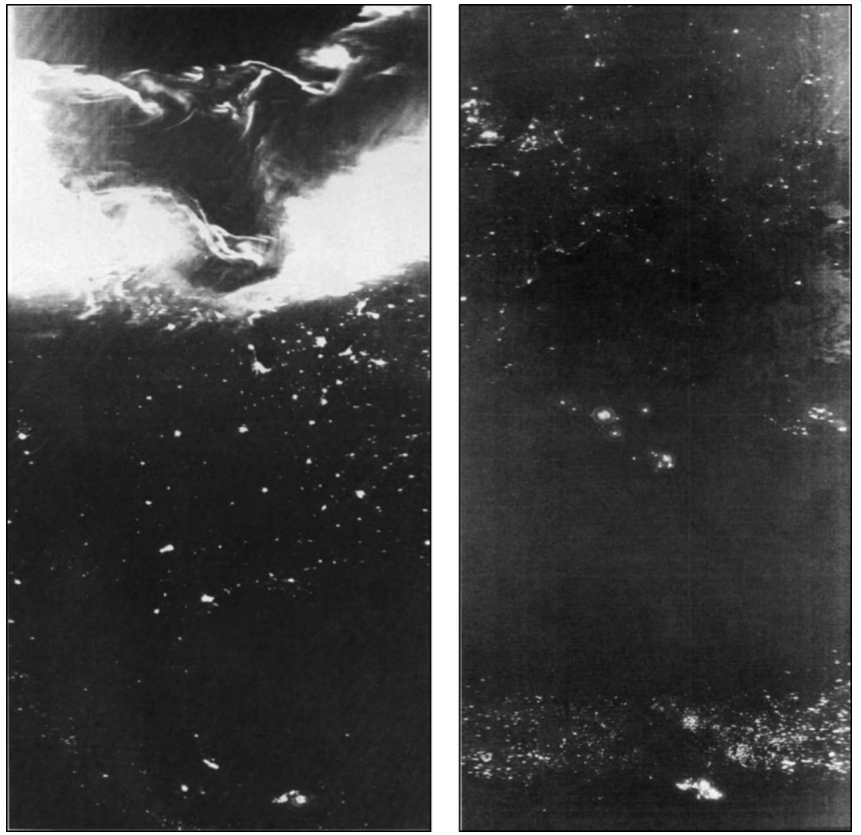
2018 — Luojia-1-01 demonstrates ≈ 130 m night imaging from a small satellite.

2021 — SDGSAT-1 Glimmer imager: colour night imagery at 10–40 m.

2022 — NOAA-21 launched: a third VIIRS extends the record into the 2030s.

The irony: the first instrument was built to see moonlit clouds. City lights were noise — it took two decades for anyone to treat the noise as the signal.

The first night images: DMSP film, 1974



Left orbit · 18 April 1974

City lights of the central United States and Mexico — and an aurora over Canada.

Right orbit · Europe to central Africa

City lights of Europe and Asia down to central Africa.

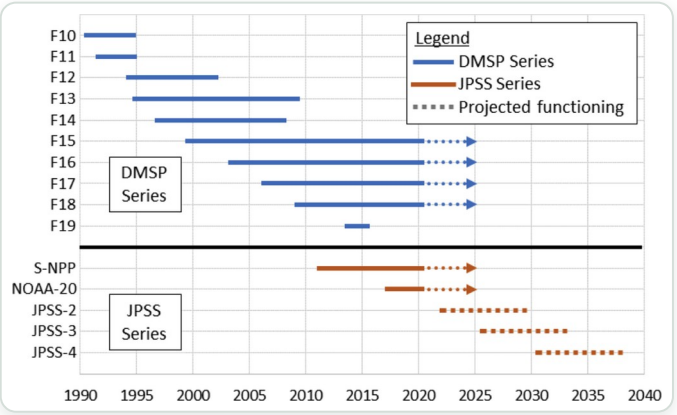
A band of fires across Africa

Agricultural burning: bright, seasonal and nothing to do with cities.

Gas flares

Nigerian offshore flares, Algeria and Libya — already visible 50 years ago.

Low-light satellites, 1990 – 2040



Why show this here? Fires and gas flares were the first African signals ever recorded from orbit — and they are still the first artefacts you must mask.

Source: Croft (1978), reproduced in World Bank Open Nighttime Lights tutorials (CC BY 4.0). Timeline: NOAA NCEI via the same source.

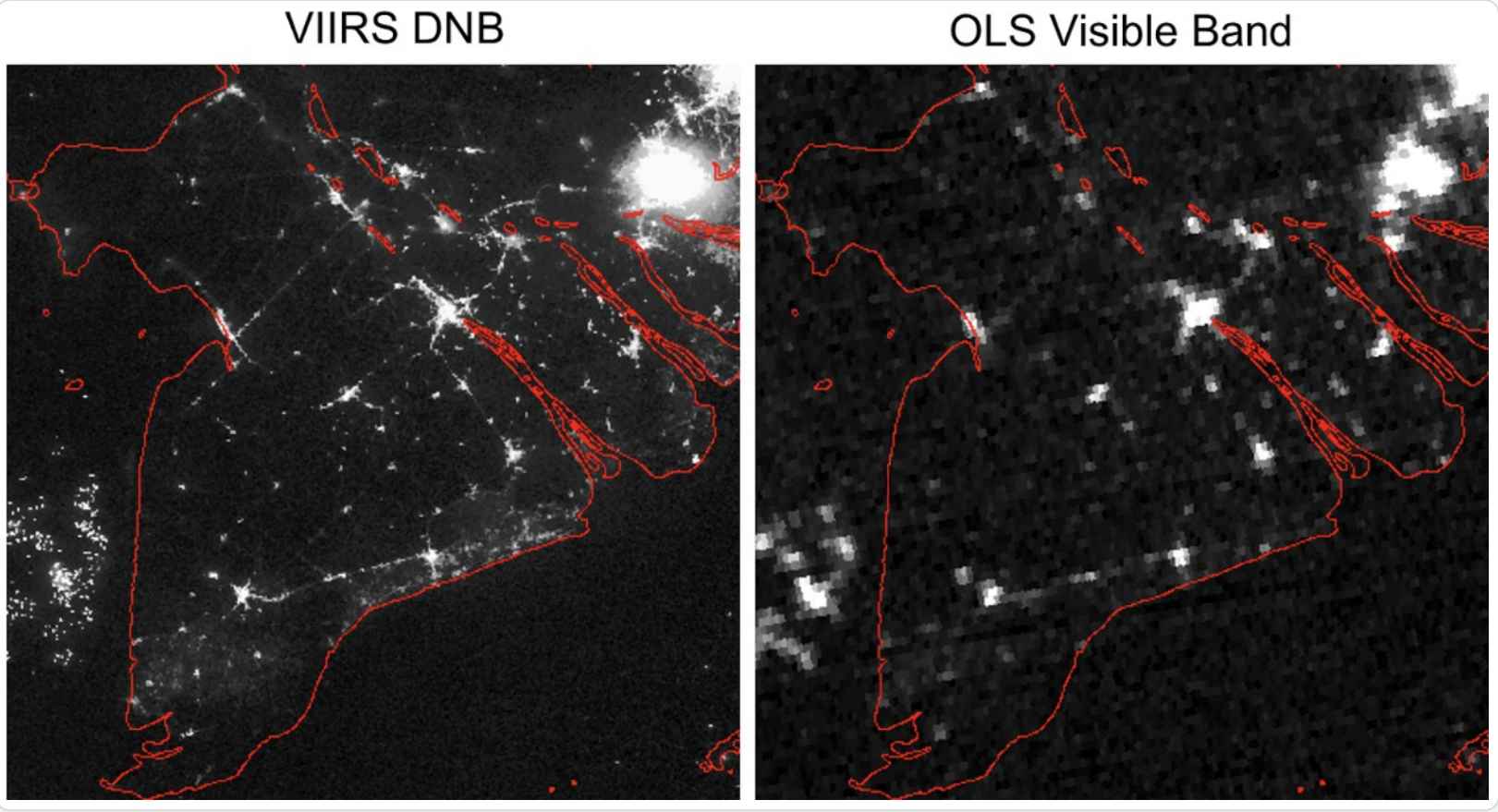
Two sensors, two different sciences

	DMSP-OLS · historical	VIIRS Day/Night Band · modern
Period	1992 – 2013 (archive)	2012 – present (standard)
Radiometry	6-bit digital numbers 0–63, no on-board calibration	14-bit, calibrated radiance in nW·cm^{-2}·sr^{-1}
Resolution	≈ 2.7 km, gridded at 30 arc-seconds (≈ 1 km)	≈ 742 m native, gridded at 15 arc-seconds (≈ 500 m)
Saturation	City cores clip at 63	Wide dynamic range, three gain stages
Sensitivity	Misses faint, rural lighting	Detects far fainter light than DMSP
Products	Annual composites only	Daily, monthly and annual — open licence

Next generation: Luojia-1-01 (≈ 130 m), SDGSAT-1 Glimmer (10 m panchromatic, 40 m colour) and commercial constellations resolve streets and can distinguish lamp types by colour — the dimension both legacy sensors lack.

REAL IMAGERY

Same night, two sensors: Mekong Delta, 23 November 2014



Finer detail

VIIRS (left) resolves individual towns, canals and roads; DMSP (right) blurs them into blobs.

Higher sensitivity

Faint rural lights along the waterways appear only in VIIRS — the lights that matter most in rural Africa.

Less blooming

DMSP's large footprint spreads light beyond its source and overstates lit area.

Take-away: for anything after 2012, use VIIRS; DMSP is for history only, and only after harmonisation.

Source: Elvidge et al. (2013), reproduced in World Bank Open Nighttime Lights tutorials (CC BY 4.0).

Why the brightest places say the least

DMSP returns 63 above a certain brightness — a suburb and a downtown core look identical.

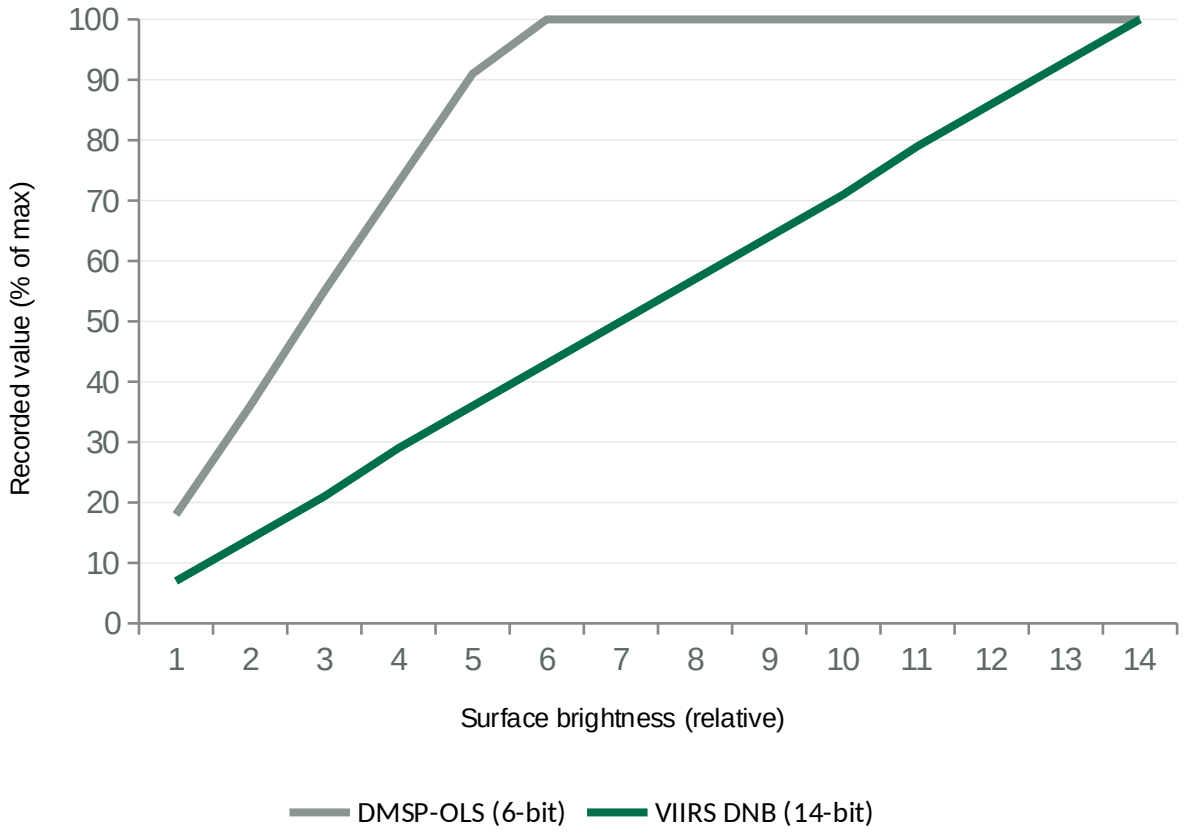
VIIRS spans roughly seven orders of magnitude, with three switchable gain stages.

Its low noise floor makes a single lit fishing boat visible on a dark lake or ocean.

Read this before you regress

Saturation is a ceiling, not noise you can average away. The information was never recorded — correcting it means modelling what the sensor failed to see.

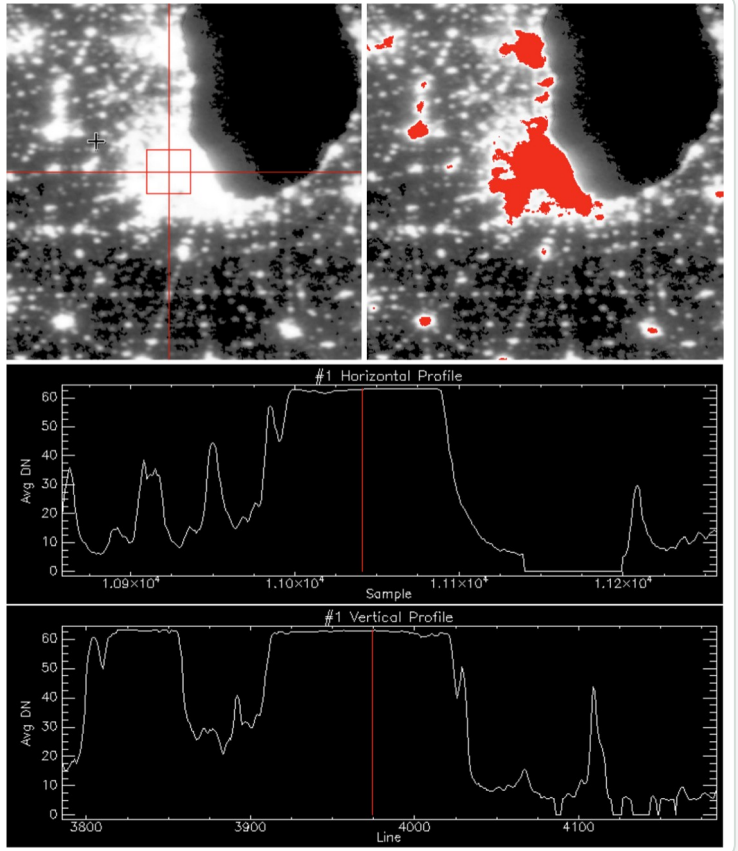
Sensor response as surface brightness rises (schematic)



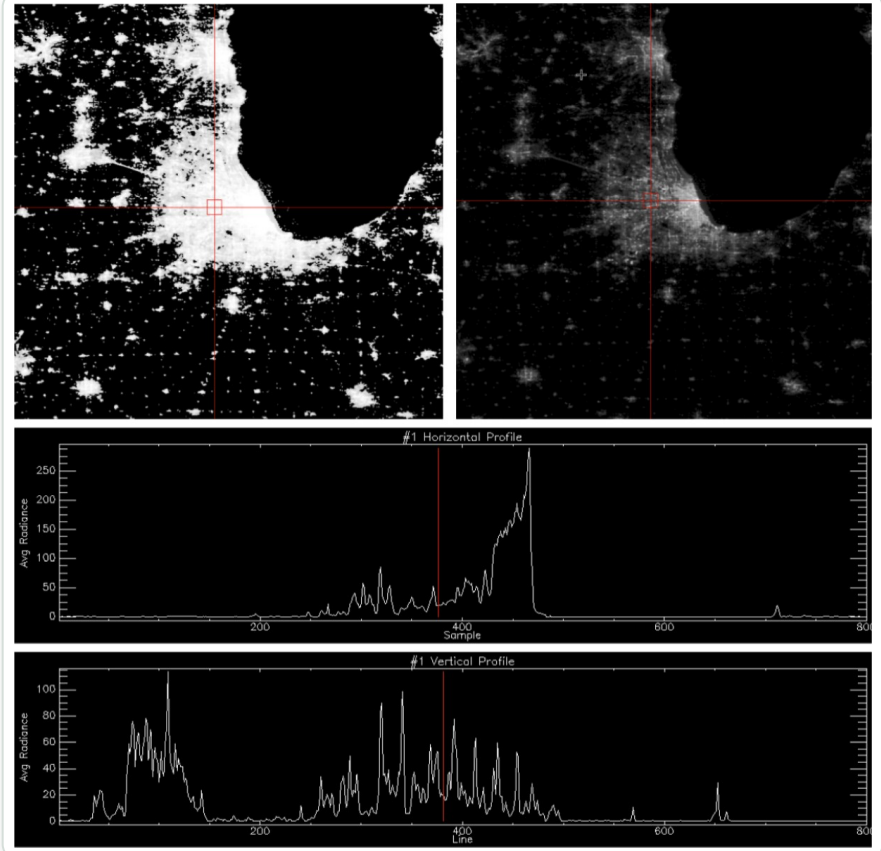
Schematic illustration — not fitted data. VIIRS range per Elvidge et al. (2017).

Saturation on real data: the same city, 2013 vs 2016

DMSP-OLS · 2013 · saturated pixels in red



VIIRS-DNB · 2016 · the same area



Read the profiles

- **DMSP:** transects flatten at 63 across the whole metro area — no information left in the core.
- **VIIRS:** the same transects keep rising and falling across the city.
- **For Africa:** Lagos, Cairo and Johannesburg cores behave like this in DMSP; rural lights sit at the other end of the range.

Chicago region. DMSP-OLS F18 2013 stable lights vs VIIRS-DNB 2016 composite. Source: World Bank Open Nighttime Lights tutorials (CC BY 4.0).

When and how VIIRS looks at your country

≈ 01:30

local equator-crossing time of the night pass (sun-synchronous orbit)

≈ 3,000 km

swath width — every point on Earth is seen at least once a night

≈ 742 m

native DNB pixel, held near-constant from nadir to edge of scan

3

VIIRS instruments in orbit: Suomi NPP, NOAA-20, NOAA-21

View angle changes every night

The same place is seen from directly above one night and obliquely the next. Tall buildings, trees and slopes block or reveal light differently, so daily radiance fluctuates for purely geometric reasons.

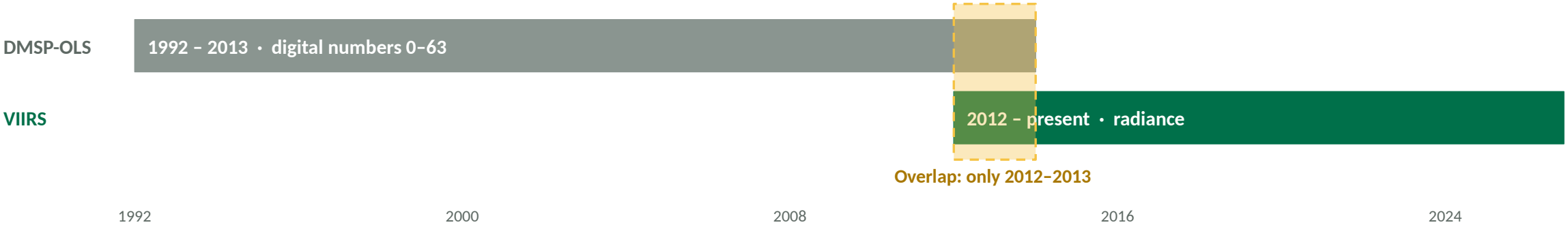
The Moon changes the scene

Moonlight reflected from bright surfaces (sand, salt pans, cloud tops) adds radiance that must be modelled out — the lunar cycle leaves a ≈ 29.5-day rhythm in uncorrected data.

Aggregate before you interpret

Composites average many view angles and lunar phases. Use daily data only when the question needs it, and smooth it (7-day or monthly) before reading a trend.

Bridging the sensors: building a long series honestly



1 VIIRS only (2012 →)

Cleanest choice for most macro questions: calibrated, consistent, monthly. Accept a shorter history.

Recommended default

2 A published harmonised product

e.g. Li et al. (2020) DMSP-like series 1992–2018; Chen et al. (2021) VIIRS-like series 2000–2018. These are model outputs — cite them as estimates.

For long-run questions

3 Splice growth rates yourself

Use DMSP growth before 2013 and VIIRS growth after, linked at the overlap. Document the method and test it against an alternative.

Only with documentation

Li, X. et al. (2020) Scientific Data 7; Chen, Z. et al. (2021) Earth System Science Data 13; Elvidge, C. D. et al. (2021) Remote Sensing 13(5).

SECTION 3

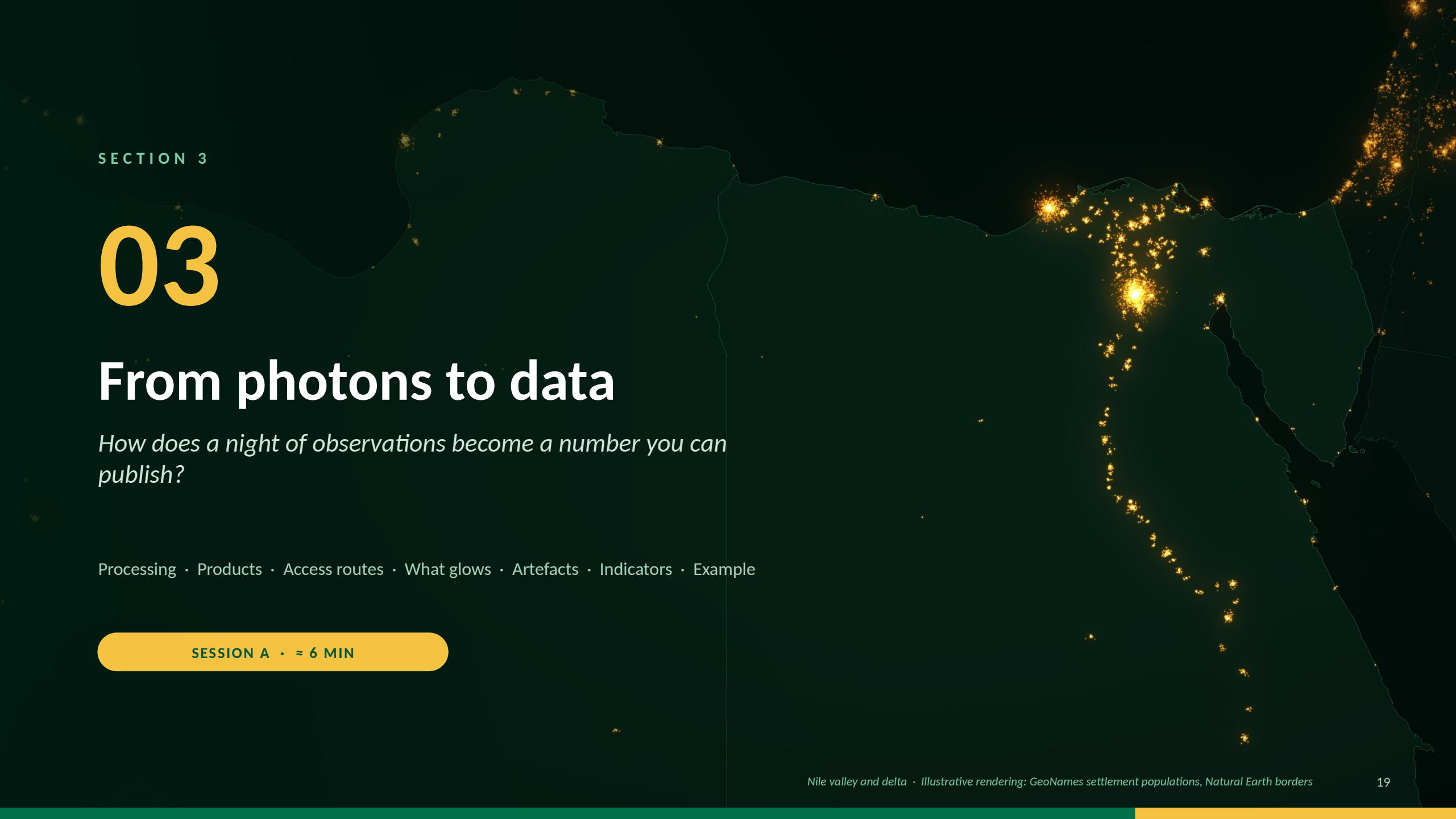
03

From photons to data

How does a night of observations become a number you can publish?

Processing · Products · Access routes · What glows · Artefacts · Indicators · Example

SESSION A · ≈ 6 MIN



Nile valley and delta · Illustrative rendering: GeoNames settlement populations, Natural Earth borders

How a night-lights image is manufactured

1

Screen for cloud

Only nights flagged cloud-free enter a composite. In the wet tropics most nights may be discarded.

2

Remove sunlight

Twilight, glint and instrument stray light are corrected or excluded.

3

Correct for the Moon

Moonlight reflected by the ground mimics emitted light; it is modelled out.

4

Filter transients

Fires, lightning, aurora and moving boats are flagged by their instability.

5

Composite & tile

Surviving nights are averaged into a monthly or annual grid, tiled globally.

You will meet these steps as layers in Lab A1

Black Marble VNP46A2 carries the corrected radiance plus its evidence: cloud-mask flag, lunar irradiance, snow flag and a Mandatory_Quality_Flag (0 = high-quality persistent, 1 = high-quality ephemeral, 2 = poor quality, 255 = no retrieval).

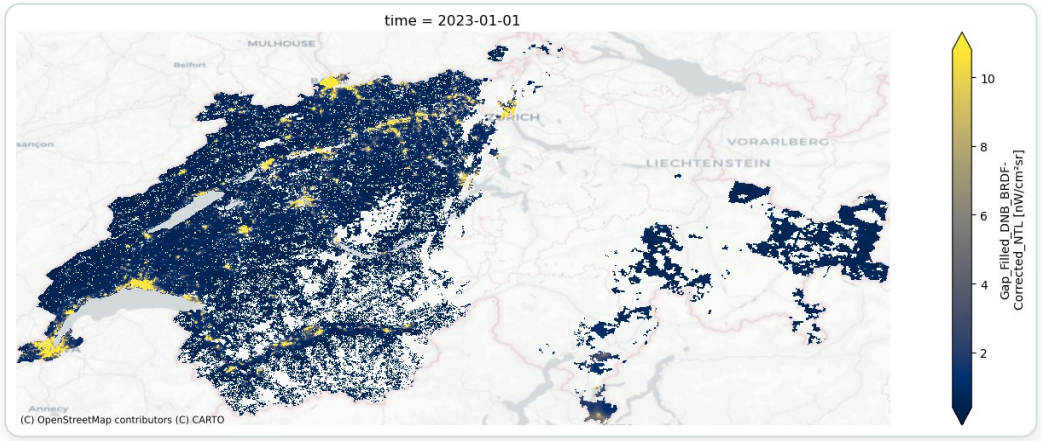
A product, not a snapshot

Two products for the same month can legitimately differ. Always record the product, collection and version.

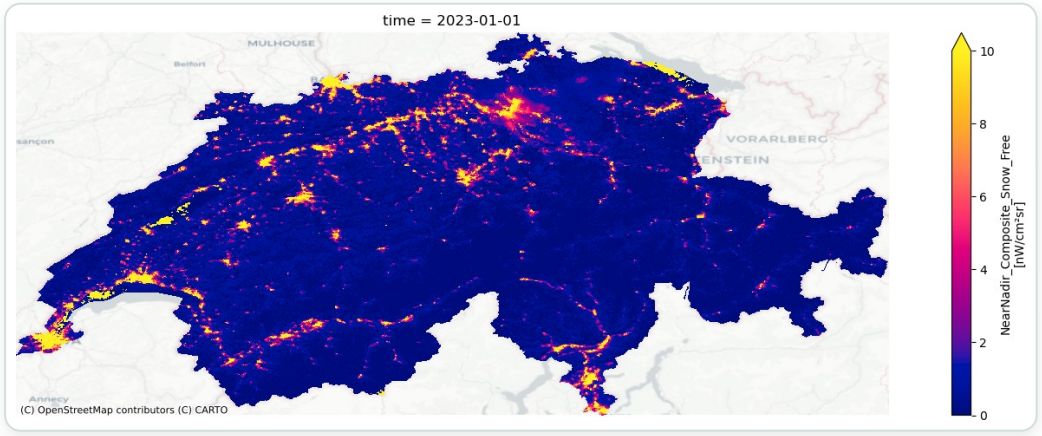
Sources: Román et al. (2018); NASA LAADS DAAC VNP46A2 collection 2 product page.

One night versus a composite: why quality flags matter

VNP46A2 · one night, poor-quality pixels removed



VNP46A3 · monthly composite, same area



Gaps are clouds, not darkness

Masked pixels (white, top) are missing data — never treat them as zero light.

Composites fill the gaps

Averaging the valid nights of a month gives a complete, smoother map (bottom).

Choose your flag threshold

Keeping only high-quality pixels is cleaner but leaves fewer valid nights — test both.

In Africa the gaps come from rainy-season cloud rather than snow — the lesson is the same: missing is not dark.

Switzerland, January 2023. Source: BlackMarblePy quality-assessment notebook (World Bank, MIT licence); NASA Black Marble; basemap © OpenStreetMap contributors © CARTO.

REFERENCE

The NASA Black Marble product family

USED IN LAB A

	VNP46A1 DAILY	VNP46A2 DAILY	VNP46A3 MONTHLY	VNP46A4 ANNUAL
Content	At-sensor radiance (top of atmosphere)	Moonlight-adjusted, gap-filled NTL	**Composite of corrected daily NTL**	Composite of corrected daily NTL
Corrections	Uncorrected — inputs & geometry only	Cloud, lunar, atmosphere, terrain, stray light	**As A2, plus number of observations & quality**	As A2, plus number of observations & quality
Typical use	Advanced users	Blackouts, reliability, events	**Programme evaluation, seasonality, nowcasting**	Long-run trends, subnational panels

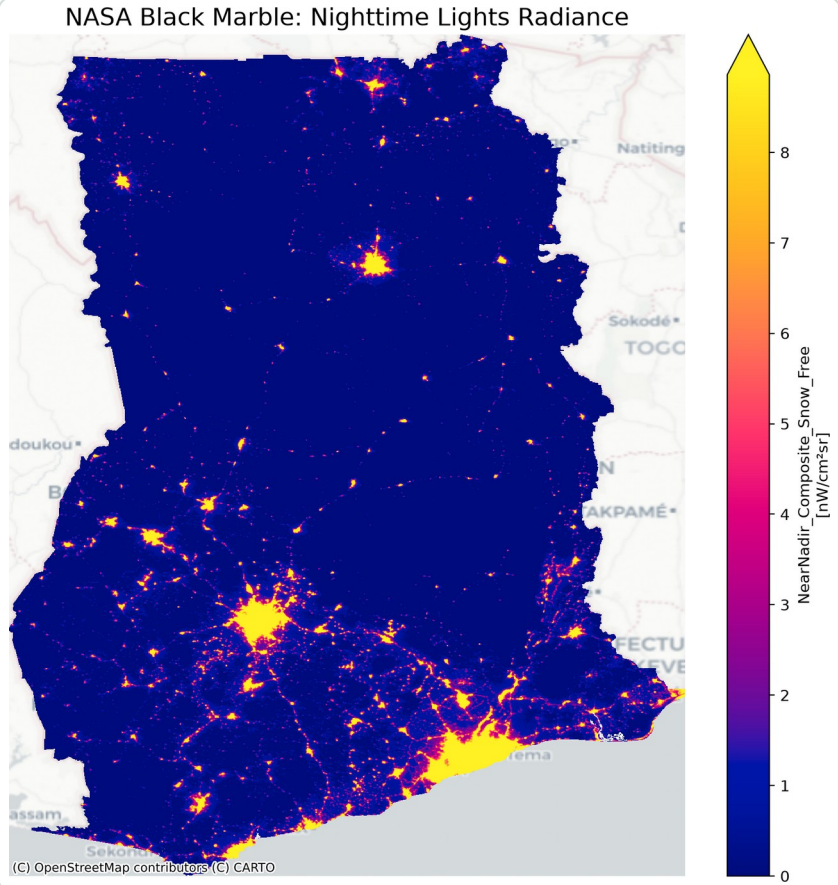
Common to all: 15 arc-second grid (≈ 500 m at the equator) · 10° × 10° tiles · HDF5 · from January 2012 · NOAA-20 twins named VJ146*.

Gotcha: radiance is stored as scaled integers — apply the scale factor in the file metadata and mask the fill value (65535 for radiance, 255 for quality). Tools such as BlackMarblePy do this for you; raw HDF5 readers do not.

Variables: A2 → DNB_BRDF-Corrected_NTL · A3 / A4 → NearNadir_Composite_Snow_Free (also OffNadir_ and AllAngle_ variants).

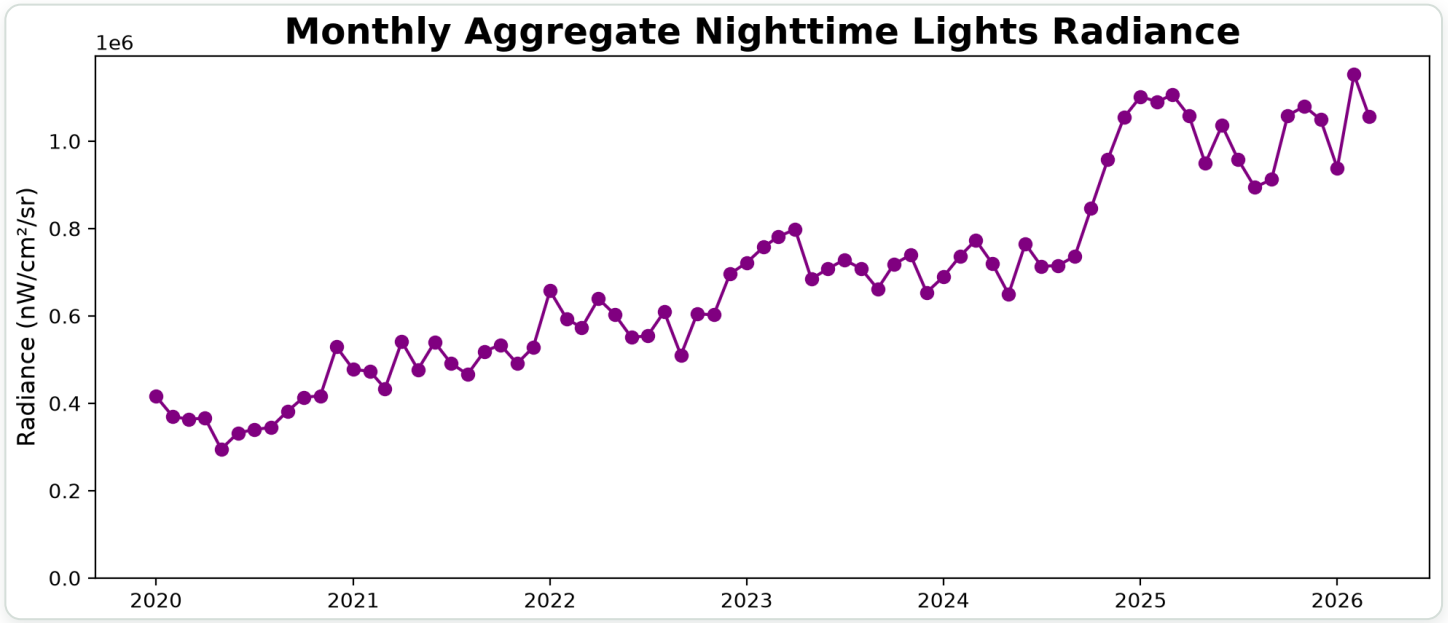
REAL IMAGERY

What Black Marble gives you: Ghana



Annual composite, VNP46A4 — Accra, Kumasi, Tamale and the corridors between them

Monthly national Sum of Lights, VNP46A3, January 2020 onwards



Lab A1 output

This is the map you will produce for your own country: clipped, in radiance units.

Lab A2 output

And this is the series: one point per month, ready for masks, quarters and growth rates.

Source: NASA Black Marble VNP46A4 (annual) and VNP46A3 (monthly), produced with BlackMarblePy (World Bank, MIT licence); basemap © OpenStreetMap contributors © CARTO.

REFERENCE

Five ways to get the data — pick one and stick to it

NASA LAADS DAAC / Earthdata	Black Marble VNP46 & VJ146 tiles in HDF5. Free Earthdata login and download token.	PRIMARY SOURCE
BlackMarblePy (World Bank)	Python: download, mosaic, clip to your boundary and compute zonal statistics in a few lines. R twin: blackmarbler.	USED IN LAB A
Google Earth Engine	Hosted catalogue: NOAA/VIIRS/DNB/MONTHLY_V1/VCMSLCFG · NOAA/VIIRS/DNB/ANNUAL_V21 & V22 · Black Marble collections.	FALLBACK
Earth Observation Group (EOG)	VIIRS Nighttime Lights (VNL) annual V2.x and monthly GeoTIFFs; VIIRS Nightfire for gas flares.	ANNUAL SERIES
World Bank Light Every Night	Every nightly DMSP and VIIRS observation as open data on AWS, with tutorials.	ADVANCED

Choosing a route

- **Reproducible pipeline?** BlackMarblePy or blackmarbler, scripted and versioned.
- **No local compute?** Earth Engine — but record collection IDs and dates.
- **Long annual series?** EOG VNL V2 annual composites.
- **Never mix** products in one series without documenting it.

What actually glows over Africa at night

Cities & settlements

SIGNAL

Streets, buildings, markets, vehicles. The dominant and most-studied part of the signal.

Corridors, ports & mines

SIGNAL

Highways, ports, airports, mining and industrial sites: the network between the bright spots.

Gas flares

MASK

Very bright, isolated points in oil & gas regions — e.g. the Niger Delta, Algeria, Libya. Mask them.

Fishing lights

CONTEXT

Lamp-attraction fishing on Lake Victoria and Lake Tanganyika; offshore fleets around island states.

Fires

MASK

Savanna and agricultural burning: intense, short-lived and strongly seasonal.

Natural & reflected light

REMOVE

Moonlight on bright desert sand or salt pans, airglow. Not a human signal at all.

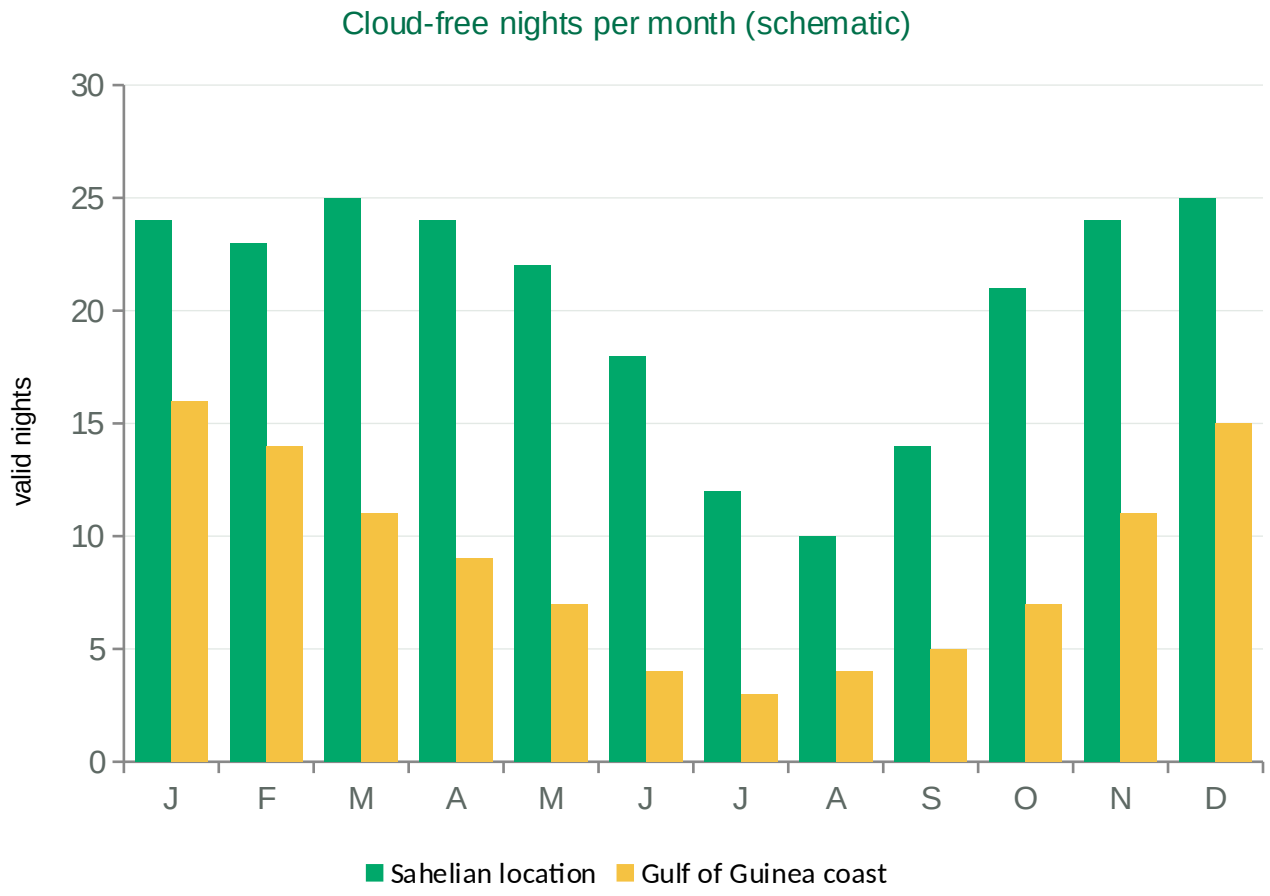
Gas flares are catalogued separately by the VIIRS Nightfire product (Earth Observation Group).

Six artefacts you must check before you trust a series

Artefact	What you see	Quick test	Treatment
Blooming / overglow	Light spreads beyond its source; small towns merge; lit area overstated	Compare lit area with a built-up footprint (GHSL, WSF)	Raise / test the threshold; use built-up masks; prefer sums at coarse boundaries
Saturation	Flat tops in city cores (DMSP); bright cores stop growing	Do maxima cluster at the sensor ceiling?	Use VIIRS; for DMSP use a harmonised or radiance-calibrated product
Gas flares	Very bright isolated points in oil & gas regions	Overlay VIIRS Nightfire flare locations	Mask flare sites with a buffer before aggregating
Seasonality	A recurring calendar pattern: festivals, fire season, rainy-season gaps	Plot each calendar month across years	Compare year-on-year; seasonally adjust before modelling
Cloud cover	Months with few valid nights; noisy or missing values	Read the number-of-observations layer	Drop months below a minimum count; gap-fill or move to quarterly
Sensor discontinuity	Jumps at DMSP→VIIRS, product versions or satellite switches	Break test around known change dates	One product and version per series; harmonise and document

Log every artefact you find, where, and how you treated it — it becomes the core of the limitations note you produce in Lab A2.

Clouds, seasons and calendars: Africa's special problem



Rainy seasons thin the record

Coastal and equatorial regions can have only a handful of valid nights in peak rainy months — the monthly value rests on very little data.

Dust and smoke dim the signal

Harmattan dust and dry-season biomass-burning smoke attenuate light even on 'clear' nights.

Calendars move the lights

Ramadan, end-of-year festivities and harvest periods change night-time behaviour; holidays measurably shift radiance.

Schematic profiles for illustration only — read the actual number-of-observations layer for your country. Calendar effects: Román & Stokes (2015), Earth's Future 3(6).

Turning an image into a number

Toy district · radiance per pixel (nW·cm^{-2}·sr^{-1})

0.1	0.3	0.2	0.0
0.4	2.5	6.0	0.6
0.2	4.0	12.0	1.5
0.0	0.8	1.2	0.3

Shaded blue = above threshold 0.5

30.1

Sum of Lights (SOL)

Total radiance inside a boundary. Workhorse of the economics literature; most sensitive to background noise.

8 px

Lit area

Pixels above a chosen threshold. Intuitive — but the threshold is a researcher choice.

3.6

Mean radiance (lit)

Brightness independent of extent. Separates a district getting brighter from one getting bigger.

Raise the threshold to 1.0 → lit area falls from 8 to 6 pixels (−25%). Always report product, version, boundary, threshold and masks — and test the threshold.

Illustrative values. Light concentration (Gini of radiance) is a fourth family, used for spatial inequality.

Normalise and aggregate deliberately

Across space

- **Boundary:** admin-1 for macro comparisons; settlement footprints for programmes.
- **Statistic:** sum for ‘how much’, mean for ‘how bright’, lit area for ‘how big’.
- **Zonal tool:** exactextract weights partial pixels at boundaries.

Normalisation

- **Per km²:** compares intensity across units of different size.
- **Per capita:** only with an independent population source (census).
- **Logs:** log(SOL) tames skew; set a noise floor before taking logs.

Across time

- **Monthly → quarterly:** average the months to match quarterly GDP.
- **Growth:** year-on-year change in the same month removes most seasonality.
- **Gaps:** never average a missing month as zero.

Workhorse transformation for macro work: $g(r, t) = \log SOL(r, t) - \log SOL(r, t - 12)$ — year-on-year growth in monthly Sum of Lights for region r.

One question, end to end

Did the 2023 mini-grid programme actually bring light to its target villages?

1 State the claim in light terms

Electrified villages should show more nights above their background radiance after commissioning.

2 Choose the product

VNP46A3 monthly to date the change; VNP46A2 daily if you also want reliability. Record the version.

3 Define the units

Settlement footprints plus a small buffer — not whole districts, where a few lit villages vanish.

4 Pick indicator & control

Mean radiance and share of lit nights; controls = similar villages outside the programme.

5 Read the difference

A step change in programme villages that is absent in controls is evidence; a common rise is regional.

6 Try to break it

Check fires, flares, moon and season; compare with utility connection or programme monitoring records.

HANDS-ON · LAB A

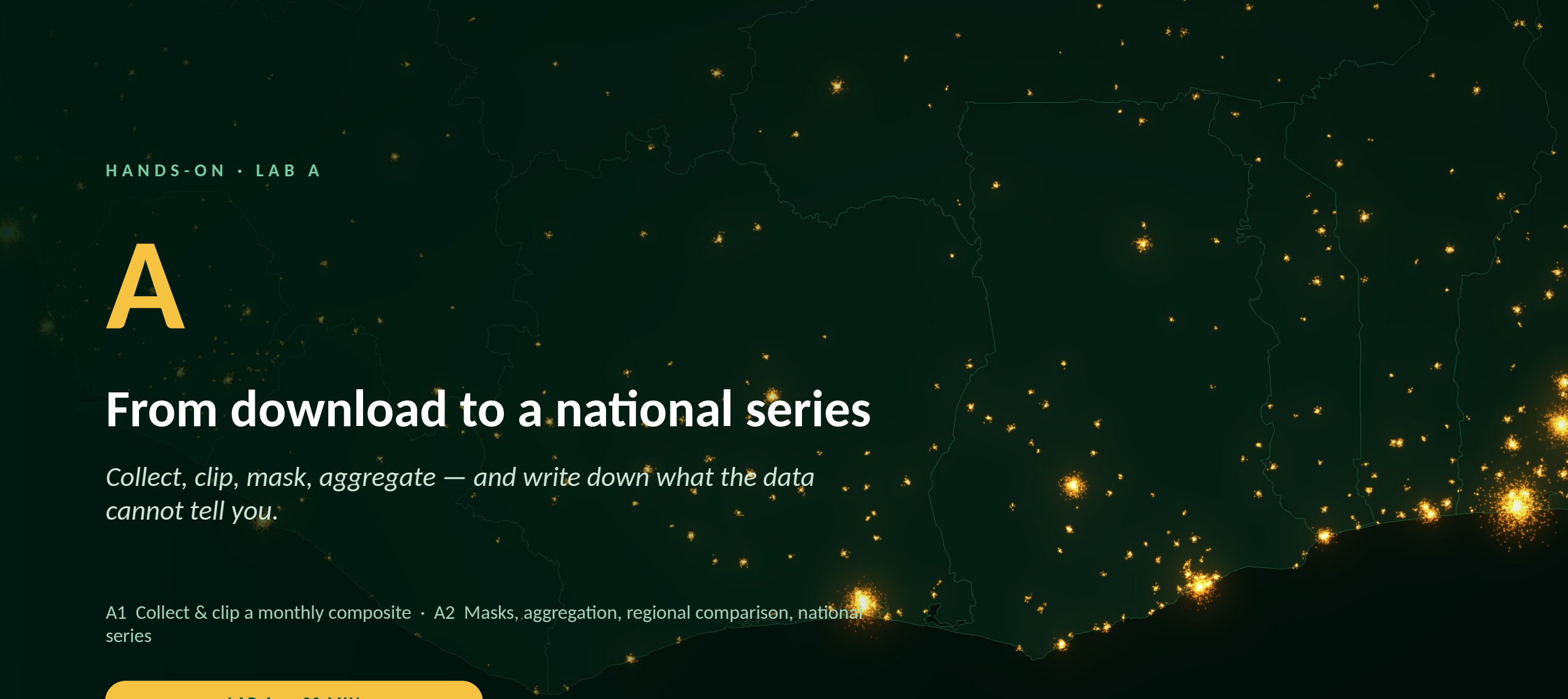
A

From download to a national series

Collect, clip, mask, aggregate — and write down what the data cannot tell you.

A1 Collect & clip a monthly composite · A2 Masks, aggregation, regional comparison, national series

LAB A · 30 MIN



Gulf of Guinea: Abidjan – Accra – Lagos · Illustrative rendering: GeoNames settlement populations, Natural Earth borders

Collect Black Marble data and clip a monthly composite

1

Get access

Create a free NASA Earthdata account and generate a download token.

2

Pick a boundary

National and admin-1 polygons: your NSO's official boundaries, or geoBoundaries / GADM.

3

Request VNP46A3

Monthly composites, January 2019 to the latest month available.

4

Clip & inspect

Map one month. Do cities, corridors — and flares or fishing lights — appear where you expect?

PYTHON · BLACKMARBLEPY

```
# pip install blackmarblepy geopandas
import geopandas as gpd, pandas as pd
from blackmarble import BlackMarble

bm  = BlackMarble(token=EARTHDATA_TOKEN,
                  output_directory="data")
gdf = gpd.read_file("boundaries/adm1.geojson")

# Monthly composites, Jan 2019 onwards
months = pd.date_range("2019-01-01", "2025-12-01",
                       freq="MS")
ntl = bm.raster(gdf, product_id="VNP46A3",
                date_range=months)
ntl = ntl["NearNadir_Composite_Snow_Free"]

# Inspect one month before going further
ntl.isel(time=-1).plot(robust=True)
```

Code is illustrative — check the BlackMarblePy documentation for the version installed in your environment.

Apply masks, aggregate, compare regions, build the series

PYTHON · MASK → AGGREGATE → COMPARE

```python
# 1  Mask: noise floor and gas-flare buffers
ntl = ntl.where(ntl >= 0.5)        # noise floor
ntl = ntl.where(~flare_mask)       # Nightfire buffer

# 2  Zonal sums by admin-1, per month
df = zonal_sum(ntl, gdf, id="ADM1") # notebook helper

# 3  Aggregate & transform
nat = df.groupby("date")["ntl_sum"].sum()
q   = nat.resample("QS").mean()
yoy = np.log(nat).diff(12)

# 4  Compare regions: share of national light
share = df.pivot(index="date", columns="ADM1",
                 values="ntl_sum")
share = share.div(share.sum(axis=1), axis=0)
```

Masks applied

Quality flags, noise floor, flare buffers, water bodies for fishing lights.

National series

Monthly SOL, quarterly average and year-on-year growth.

Regional comparison

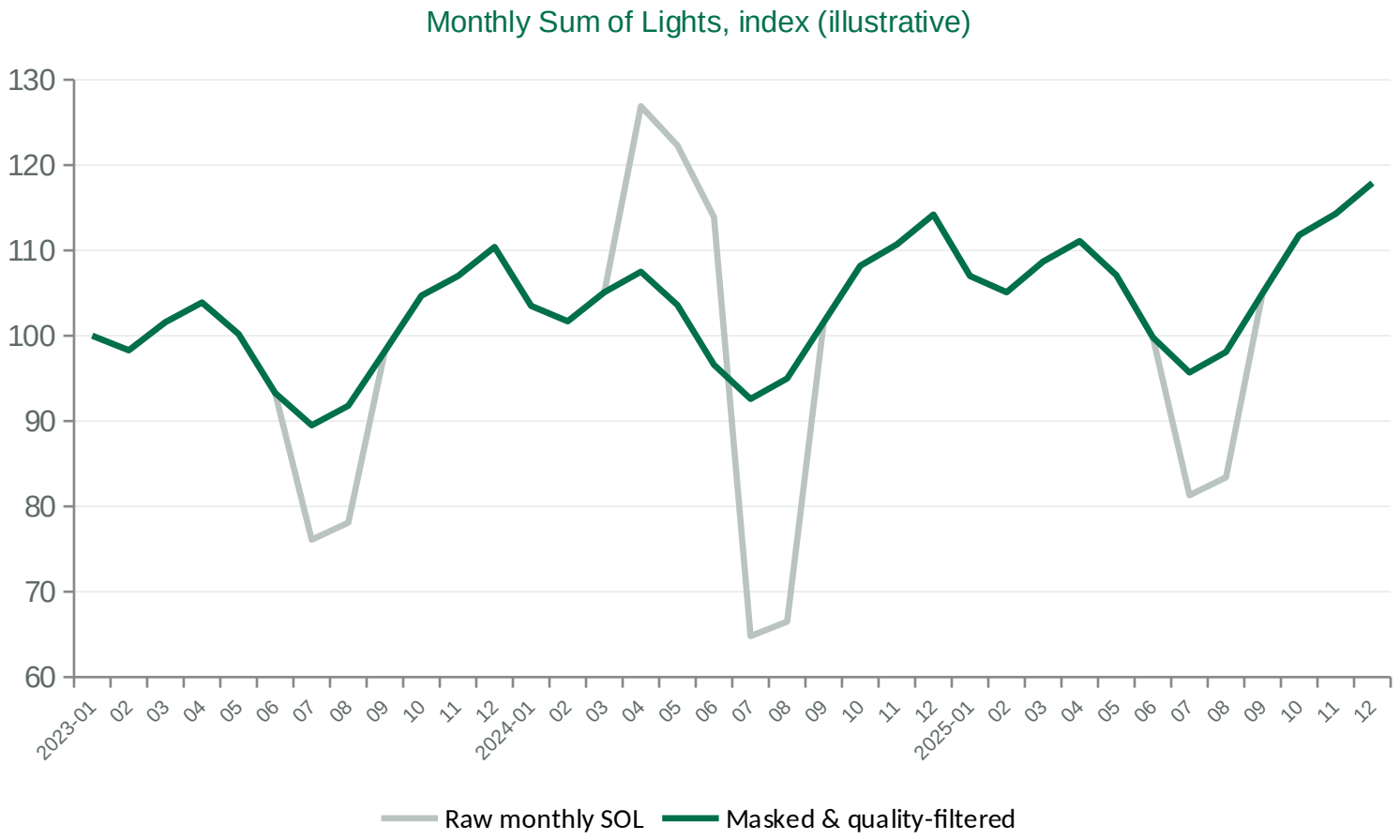
Each region's share of national light, over time.

Limitations note

Five lines: product & version, masks, threshold, cloud-poor months, known breaks.

zonal_sum wraps exactextract in the lab notebook; print df.head() before aggregating. Threshold 0.5 nW·cm^{-2}·sr^{-1} is a starting point to be tested.

A national series you can defend



LIMITATIONS NOTE — EXAMPLE

- **Data:** VNP46A3, collection 2, Jan 2023 – Dec 2025, near-nadir snow-free.
- **Masks:** quality flags; noise floor 0.5; 2 km flare buffers; inland water.
- **Gaps:** July–August have < 5 valid nights in coastal provinces; flagged, not imputed.
- **Breaks:** flare episode Apr–Jun 2024 removed by mask; no product change.
- **Do not use for:** levels of GDP, household income, small-settlement counts.

Illustrative synthetic data — shows the shape of the expected output, not any real country.

When the numbers look wrong: symptom, cause, fix

Symptom	Likely cause	Fix
Values ten times too large, or huge spikes	Scale factor not applied; fill value (65535) read as radiance	Use BlackMarblePy, or apply scale and mask fill manually
Straight seam across the country	Tile boundary — one 10° tile missing for that month	Check every tile downloaded; re-request the missing one
Boundary and raster do not overlap	CRS mismatch (projected boundary vs geographic raster)	Reproject the boundary to EPSG:4326 before clipping
Coastal districts lose light	Pixels straddling the coastline dropped by the clip	Use all-touched clipping or exactextract partial weights
One month is empty or near zero	Too few cloud-free nights; whole month gap	Flag as missing; never average it as zero
Sudden jump in a single region	Gas flare, fire, fishing fleet — or a real event	Map the month; check Nightfire and news before believing it

Keep a pre-staged copy of VNP46A3 for your countries in case downloads stall.

SECTION 4

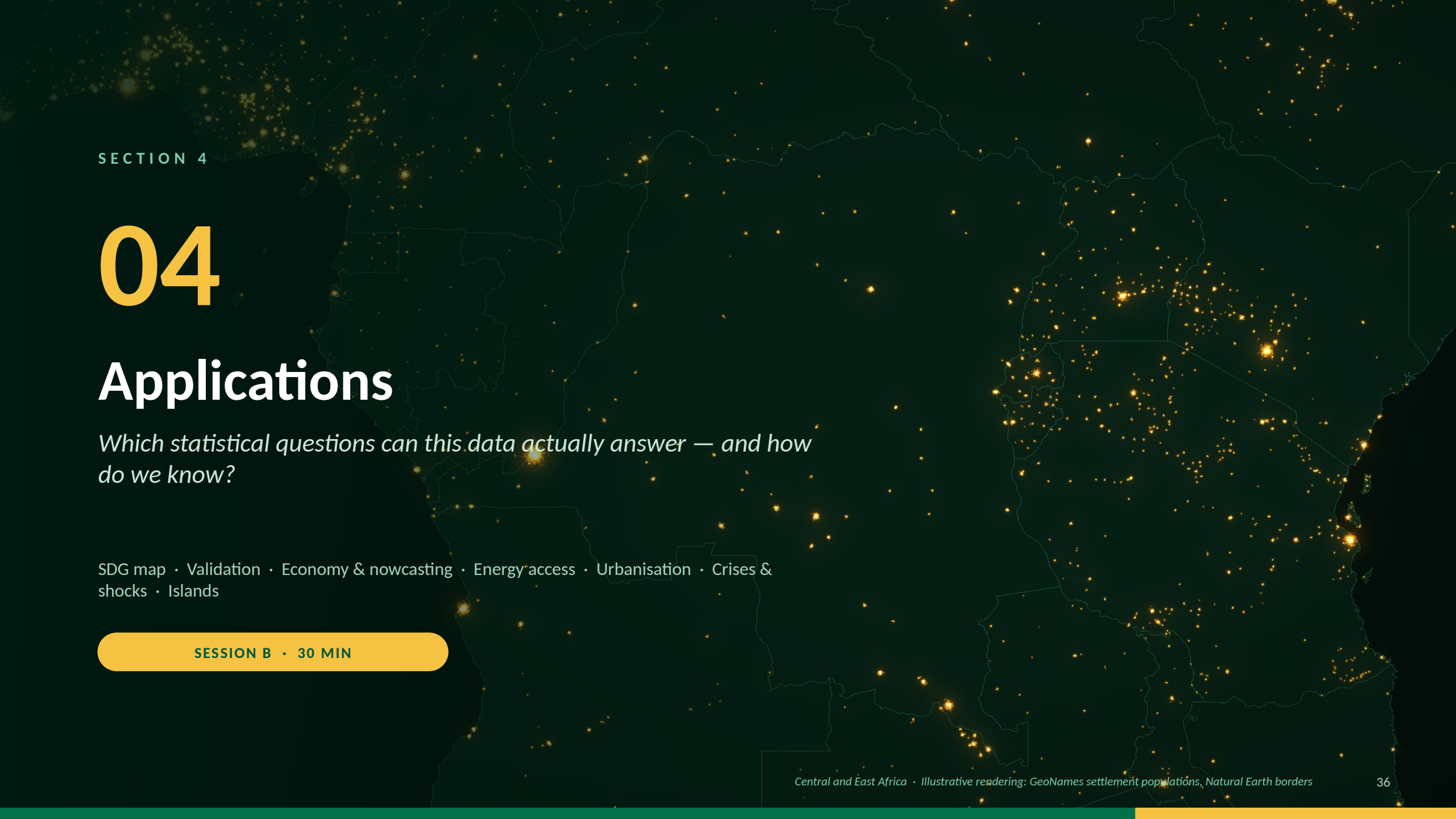
04

Applications

Which statistical questions can this data actually answer — and how do we know?

SDG map · Validation · Economy & nowcasting · Energy access · Urbanisation · Crises & shocks · Islands

SESSION B · 30 MIN



Central and East Africa · Illustrative rendering: GeoNames settlement populations, Natural Earth borders

Where lights can support SDG and macro monitoring

SDG 1

Poverty & wealth mapping

Small-area estimates combining lights with survey data

SDG 7

Electricity access & reliability

Locate unserved settlements; track how often lights are on

SDG 8

Economic activity

Subnational GDP downscaling; nowcasting; short-term activity signals

SDG 9

Infrastructure

Corridors, ports, industrial and special economic zones

SDG 11

Urbanisation

Urban extent, sprawl versus densification, new towns

SDG 16 · humanitarian

Conflict & disasters

Blackouts, displacement signals, recovery tracking

A complement, not a replacement. Lights add spatial detail, frequency and independent verification to official statistics — they do not replace the census, the survey or the national accounts.

Five use cases, five validation strategies

Use case	NTL indicator	Validated against	Performs well	Performs poorly
Output proxy	Growth in Sum of Lights	National & subnational GDP; quarterly GDP	Weak statistical systems; subnational; urban & industrial economies	Agriculture- or mining-led economies; saturated rich cores
Electrification	Share of nights lit per settlement	Household surveys (DHS, LSMS), census, utility records	Grid-connected settlements; reliability over time	Solar home systems; very low consumption
Urbanisation	Lit area and radiance growth	Built-up grids (GHSL, WSF); census urban population	City extent and growth of large towns	Small towns (blooming); dense informal areas
Damage assessment	Drop vs pre-event baseline	UNOSAT and ground damage reports; outage records	Grid outages; large-scale destruction; recovery	Localised damage; areas already dark
Activity mapping	Daily / monthly radiance change	Electricity consumption, mobility, firm surveys	Lockdowns, holidays, large events	Daytime and low-light activity

Henderson et al. (2012); Beyer, Hu & Yao (2022); Falchetta et al. (2019); Min & O'Keeffe (HREA); Román et al. (2019); Beyer et al. (2021); Li & Li (2014).

Measuring the economy from orbit

What the evidence shows

Henderson, Storeygard & Weil (2012)

Lights carry real information on economic growth and can improve GDP estimates where national accounts are weak.

Bruederle & Hodler (2018)

Lights track household wealth, education and health at the local level, using DHS survey clusters in Africa.

Martinez (2022)

Lights used as an independent benchmark to test the credibility of reported GDP growth.

Why the lights–output link is not stable

- Sector mix: agriculture and mining create value with little light
- Income level: bright cores saturate or plateau
- Lighting technology: LED retrofits change light per watt
- Power supply: load-shedding dims lights, not output

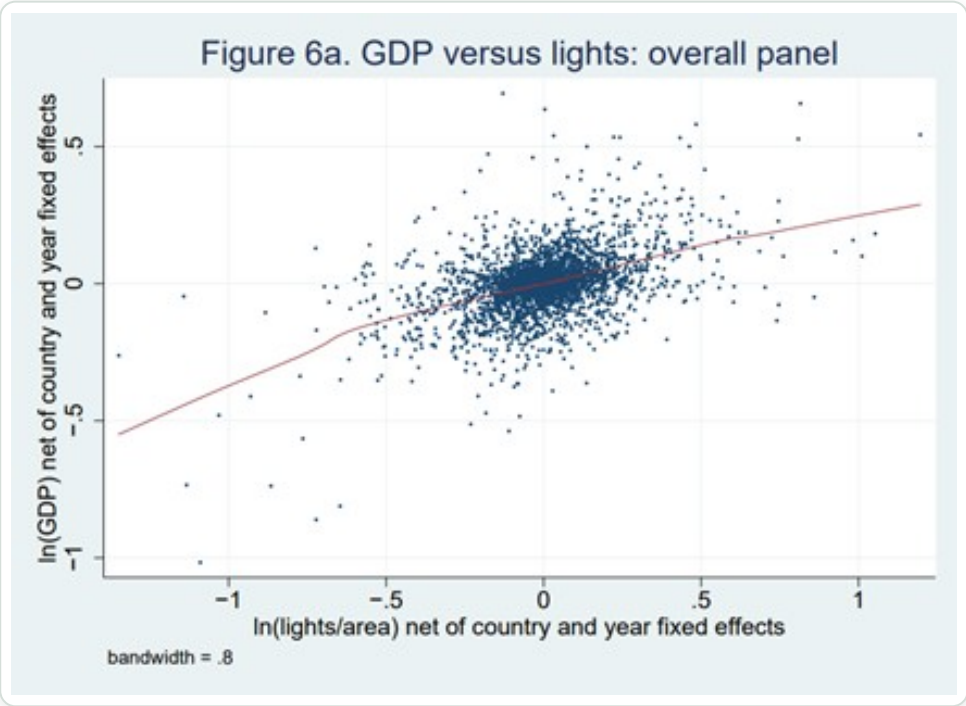
WHERE LIGHTS HELP MOST

Subnational | High-frequency | Informal economy

Henderson et al. (2012) AER 102(2); Bruederle & Hodler (2018) PLOS ONE 13(9); Martinez (2022) JPE 130(10).

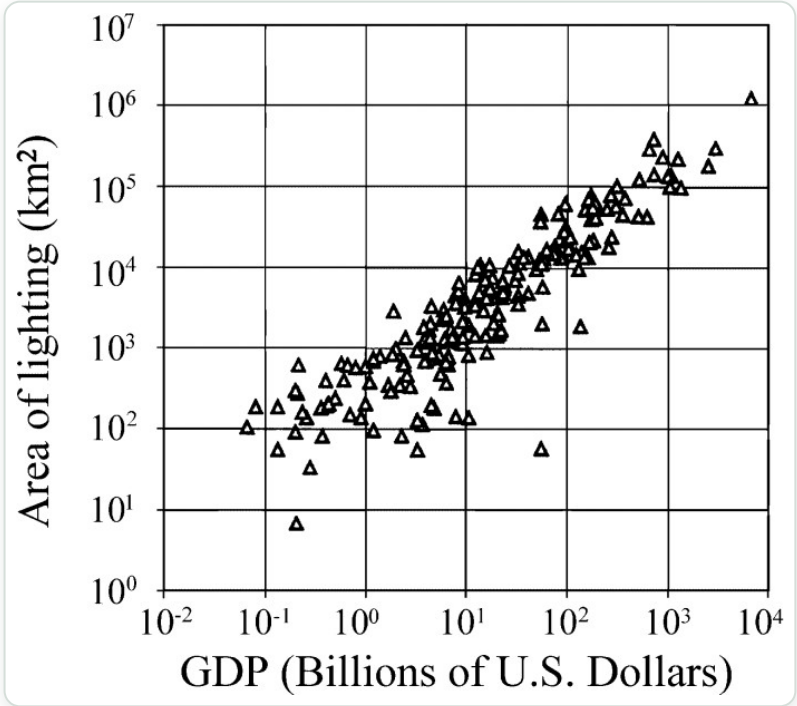
The evidence, as published

Henderson, Storeygard & Weil (2012) · GDP vs lights, within countries



Changes in lights and changes in GDP move together after removing country and year effects — but with wide scatter.

Elvidge et al. (2001) · lit area vs GDP, 200 countries



Across countries, lit area rises almost one-for-one with GDP on a log scale — the pattern that launched the field.

Henderson, Storeygard & Weil (2012), Fig. 6a; Elvidge et al. (2001). Both reproduced in World Bank Open Nighttime Lights tutorials (CC BY 4.0).

Four ways to turn lights into an output estimate

1 Elasticity regression

$$\Delta \log GDP(r,t) = \beta \cdot \Delta \log NTL(r,t) + \text{region \& year effects}$$

Estimate how strongly output growth co-moves with light growth, using regions with good data.

2 Combine with official data

$$\hat{y} = \lambda \cdot \text{official growth} + (1 - \lambda) \cdot \text{lights-predicted growth}$$

Weight the two noisy signals by their reliability — more weight on lights where accounts are weak.

3 Spatial downscaling

$$GDP(\text{cell}) = GDP(\text{region}) \times \text{NTL share(cell)}, \text{ refined with covariates}$$

Distribute known regional totals to a grid or to districts with no accounts.

4 Nowcasting

$$\text{Quarterly GDP} \leftarrow f(\text{monthly NTL growth, other indicators})$$

Use the monthly series as a timely feature before official data are released.

The elasticity is not a constant. Hu & Yao (2022) find the lights–GDP relationship is non-linear and differs by income level — estimate it for your context rather than borrowing a number.

Henderson et al. (2012); Hu & Yao (2022) Journal of Econometrics 228(2); Beyer, Hu & Yao (2022) IMF Working Paper 22/109.

Lights as a nowcasting feature — from series to model

Monthly NTL

VNP46A3, masked, national & regional

›

Transform

log, year-on-year growth, quarterly average

›

Model

bridge equation, MIDAS or machine learning

›

Nowcast

quarterly GDP growth, before release

Strengths

- Available weeks after the month ends
- Independent of surveys and administrative reporting
- Regional detail that national indicators lack

Pitfalls

- Look-ahead bias: use only data available at nowcast date
- Breaks from product changes look like recessions
- Seasonality and cloud gaps must be treated first

Evidence

Beyer, Hu & Yao (2022, IMF WP 22/109) use night-time lights to measure quarterly economic growth across countries. Beyer et al. (2021) combine daily lights with electricity consumption to track the COVID-19 shock in India.

Energy access and SDG 7: Africa's defining gap

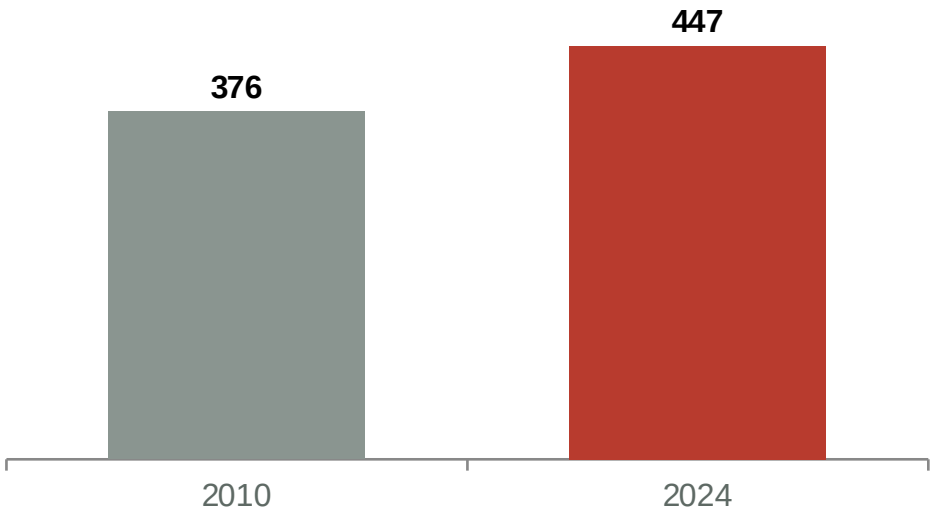
655 M

people without electricity worldwide (2024)

> 560 M

of them in sub-Saharan Africa

Rural sub-Saharan Africa without electricity (millions)



Locate the unserved

Settlements visible by day but dark at night are candidate unelectrified communities.

Verify programmes

A grid extension that produces no new light has not, in a meaningful sense, been delivered.

Measure reliability

Daily data reveal how often lights are on — load-shedding a binary "connected" statistic hides.

Plan the last mile

Distance to lit corridors informs grid extension vs mini-grid vs stand-alone solar.

Mission 300 (World Bank Group & AfDB): 300 M people connected by 2030 — monitoring at that scale needs spatial evidence.

Sources: Tracking SDG 7: The Energy Progress Report (IEA, IRENA, UN DESA, World Bank, WHO), 24 June 2026; HREA, Min & O'Keeffe (Univ. of Michigan).

Connected is not the same as powered

Thirty nights in two villages that both count as “connected” (illustrative)

Village A · stable grid



29 / 30 nights lit

Village B · load-shedding



15 / 30 nights lit

■ lit night ■ dark night

How it works: for each settlement, compare every night’s radiance with a statistical model of the dark background around it. The share of nights clearly above background is a reliability proxy; its trend shows whether access is improving.

For a statistical office

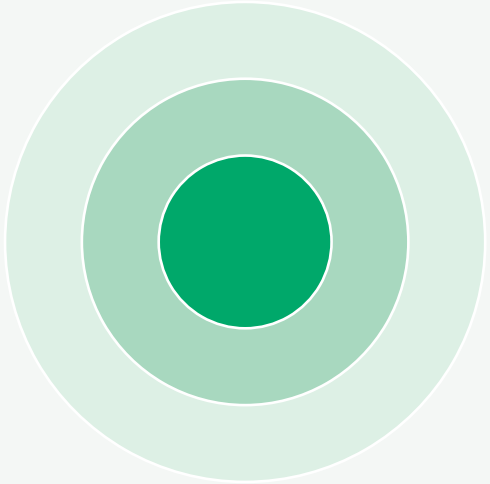
- Add a reliability column next to access rates
- Flag districts where access rose but lit nights did not
- Cross-check utility outage logs

Illustrative nights. Method in the spirit of HREA (Min & O’Keeffe, University of Michigan / World Bank); see also Falchetta et al. (2019), Scientific Data 6.

Urbanisation and the shape of a city

Sprawl

Footprint expands outward at low intensity



Lit area +40% · Radiance +6%

Densification

The same footprint intensifies



Lit area +6% · Radiance +40%

Map the footprint where registers are missing

Lights reveal built-up extent where cadastral records or recent maps do not exist.

Separate sprawl from densification

Compare growth in lit area with growth in radiance — it matters for transport and services.

Detect new towns early

Mining boomtowns, economic zones and displacement settlements appear before official geography.

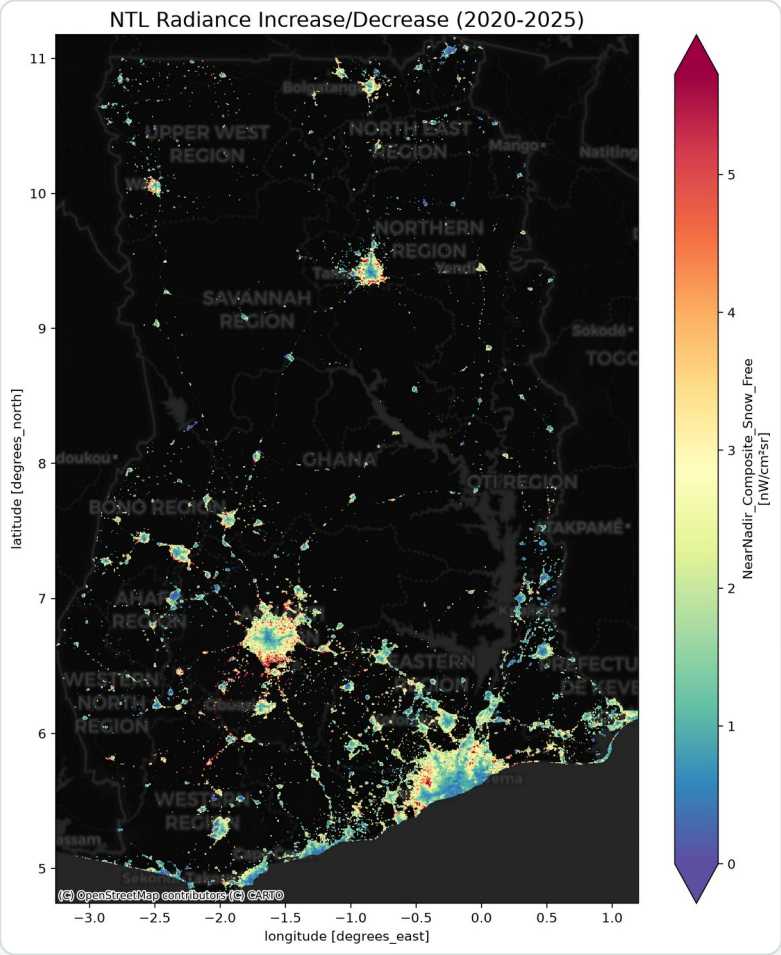
Pair with population grids

Use census-based or built-up grids — bright pixels are not people.

Same Sum of Lights — two very different cities (illustrative).

REAL IMAGERY

Where did the lights grow? Ghana, 2020 – 2025



Urban edges light up

The strongest gains ring Accra and Kumasi — expansion at the edge, not only brighter cores.

Corridors fill in

New light along the roads linking the main cities: the settlement network densifying.

New lights in the north

Small new spots in northern regions: candidate electrification gains to verify on the ground.

Test it in Lab B

Compare regional change with census or utility data — do the regions that brightened most also grow fastest?

Source: NASA Black Marble VNP46A3 change 2020–2025, produced with BlackMarblePy (World Bank, MIT licence); basemap © OpenStreetMap contributors © CARTO.

Lights are strongest in combination with other grids

Lights × built-up

Built-up grids

GHSL, World Settlement Footprint, Google Open Buildings

Built but dark = candidate unelectrified or abandoned settlements. Lit but unbuilt = flares, fires, fishing, roads.

Lights × cropland

Cropland maps

Cropland masks and crop-type maps

Explains weak lights–GDP links in agricultural regions; separates farm output from urban activity.

Lights × population

Validation slide

Census-based grids; WorldPop with care

Light per capita and unlit population. Beware circularity: WorldPop uses NTL as a covariate.

Lights × networks

Trade & ports

Roads, ports, grid lines, mines

Distinguishes corridors and enclaves from settlements; supports trade and infrastructure analysis.

When the lights go out: conflict and disaster

Conflict · Sudan, 2023

After fighting began in Khartoum in April 2023, the capital's lights fell to the noise level by May (EOG). UNOSAT mapped power outages from NTL in June 2023.

Conflict · Syria, 2011–2014

Li & Li (2014) measured a loss of more than 70% of night-time light nationally, with losses tracking displacement province by province.

Disaster · Puerto Rico, 2017

After Hurricane Maria, NASA Black Marble daily data mapped the blackout and tracked restoration neighbourhood by neighbourhood.

The interpretive rule: absence of light is not absence of people

Darkness may mean displacement, destruction, fuel shortage, curfew, a grid failure — or a decision to switch off street lights. The satellite cannot tell which. Corroboration with ground information is not optional, and uncertainty must be stated.

Sources: Payne Institute / EOG (Sept 2024); UNOSAT (June 2023); Li & Li (2014) IJRS 35(18); Román et al. (2019) PLOS ONE 14(6).

Activity mapping: shocks, events and calendars

Lockdowns

Daily lights and electricity use tracked the depth and timing of the 2020 COVID-19 shock at district level in India.

Festivals & holidays

Ramadan and end-of-year holidays measurably change night-time light — a calendar effect to model, or a signal to study.

Grid events

National blackouts and rationing episodes show up as abrupt, wide-area drops in daily data.

Cyclones & floods

Outage extent and restoration speed after storms — highly relevant for Indian Ocean economies.

Event-study recipe

1. Daily VNP46A2 for the affected area and a comparison area
2. Same calendar window in the previous year as baseline
3. 7-day rolling mean to absorb view-angle noise
4. Express as % deviation from baseline, with a band of normal variation
5. Corroborate with utility, news or field reports

Beyer, Franco-Bedoya & Galdo (2021) World Development 140; Román & Stokes (2015) Earth's Future 3(6).

An island lens: small islands and coastal economies

Few pixels

A small island state covers only thousands of 500 m cells; one bright resort, port or power plant can dominate the total.

Coast and sea mix

Coastal pixels blend land light with reflection from the water; clip with care and test all-touched versus centre-only rules.

Lights offshore

Fishing fleets, ships at anchor and offshore platforms sit just beyond the coastline — mask the sea or keep it as a separate maritime indicator.

Orographic cloud

Trade-wind cloud over central plateaus thins valid observations inland — check the observations layer by district.

Tourism signal

Hotel strips are very bright relative to their population; normalise carefully and read alongside arrivals data.

Cyclone season

Daily data can map outages and restoration after tropical cyclones — an operational use for disaster statistics.

Rule for small units: the fewer the pixels, the more each masking and boundary choice matters — report sensitivity to them explicitly.

HANDS-ON · LAB B

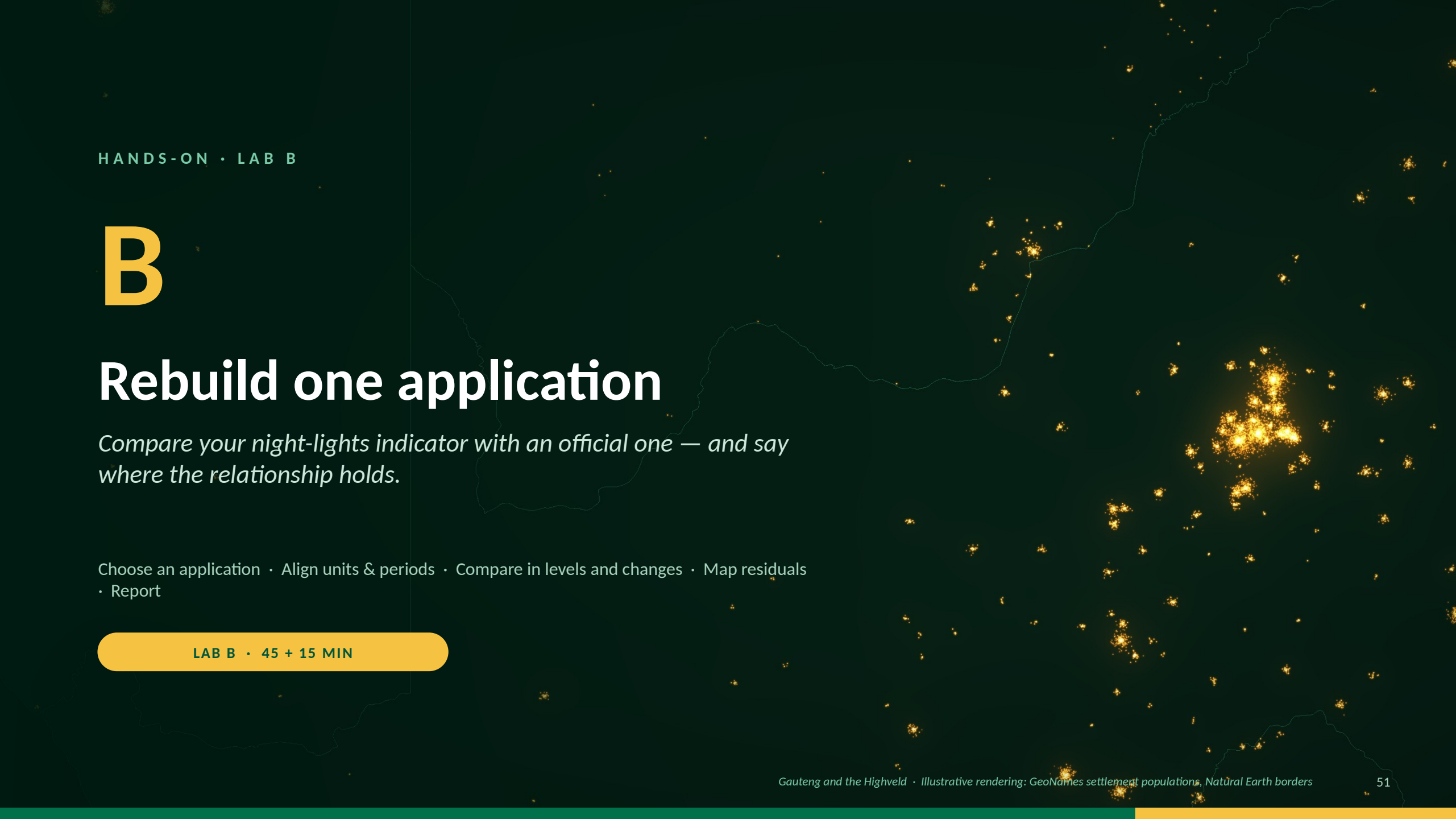
B

Rebuild one application

Compare your night-lights indicator with an official one — and say where the relationship holds.

Choose an application · Align units & periods · Compare in levels and changes · Map residuals · Report

LAB B · 45 + 15 MIN



Gauteng and the Highveld · Illustrative rendering: GeoNames settlement populations, Natural Earth borders

Pick one application and its official comparator

A

Output proxy

NTL INDICATOR

Annual SOL by region (VNP46A4 or VNP46A3 averaged)

OFFICIAL COMPARATOR

Regional GDP / GVA, or national quarterly GDP

TEST

Elasticity in growth rates; fit by region

B

Electrification

NTL INDICATOR

Share of settlements lit, by district

OFFICIAL COMPARATOR

Census or DHS share of households with electricity

TEST

Correlation across districts; outliers

C

Urbanisation

NTL INDICATOR

Lit area above threshold, by district

OFFICIAL COMPARATOR

Census urban population or built-up area

TEST

Levels and growth; sprawl vs densification

D

Shock

NTL INDICATOR

Daily or monthly radiance vs baseline

OFFICIAL COMPARATOR

Utility outage reports, disaster or activity statistics

TEST

Timing and depth of the drop; recovery

Rebuild means: reproduce the logic of a published use case on your own country, with your own official data.

Compare in levels and in changes — then map what is left

2 Align

Same units (regions or districts) and the same periods for both series.

3 Transform

Logs for skewed totals; growth rates for change; per km² or per capita if justified.

4 Compare

Scatter, correlation and slope — once in levels, once in changes.

5 Map residuals

Where does the official indicator sit far above or below what lights predict?

6 Stress-test

Drop the capital; change the threshold; remove masks one at a time.

PYTHON · OPTION A EXAMPLE

```
import numpy as np, statsmodels.formula.api as smf
d = ntl_region.merge(official, on=["region", "year"])
d["ln_ntl"], d["ln_y"] = np.log(d.sol), np.log(d.gdp)

# Levels: cross-section, pooled years
lev = smf.ols("ln_y ~ ln_ntl + C(year)", d).fit()

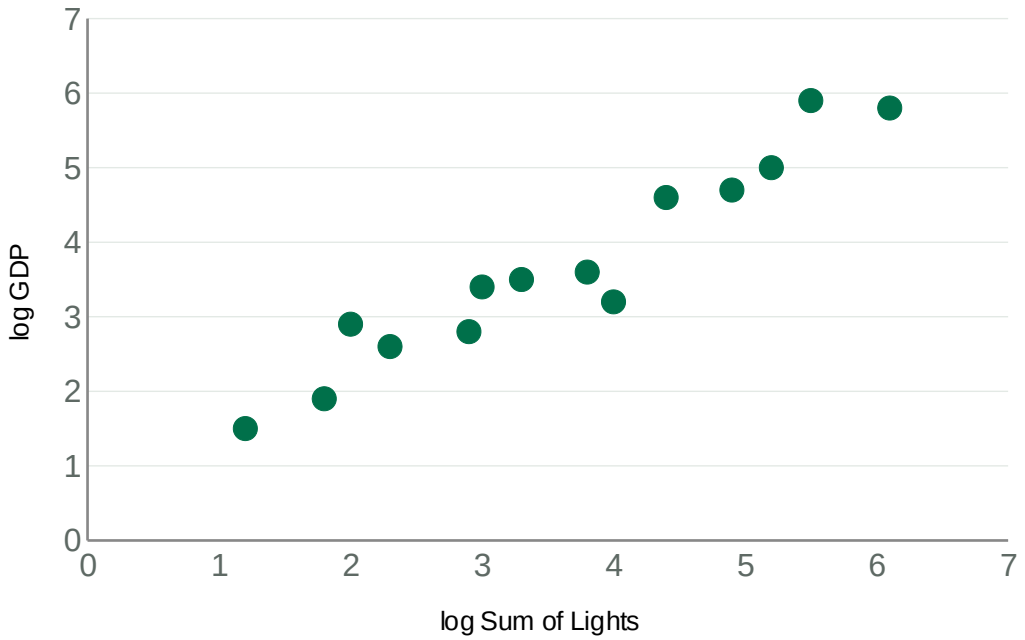
# Changes: growth within region over time
d = d.sort_values(["region", "year"])
g = d.groupby("region")[["ln_y", "ln_ntl"]].diff()
chg = smf.ols("ln_y ~ ln_ntl", g.dropna()).fit()

print(lev.params, lev.rsquared)
print(chg.params, chg.rsquared)
d["resid"] = lev.resid      # map this
```

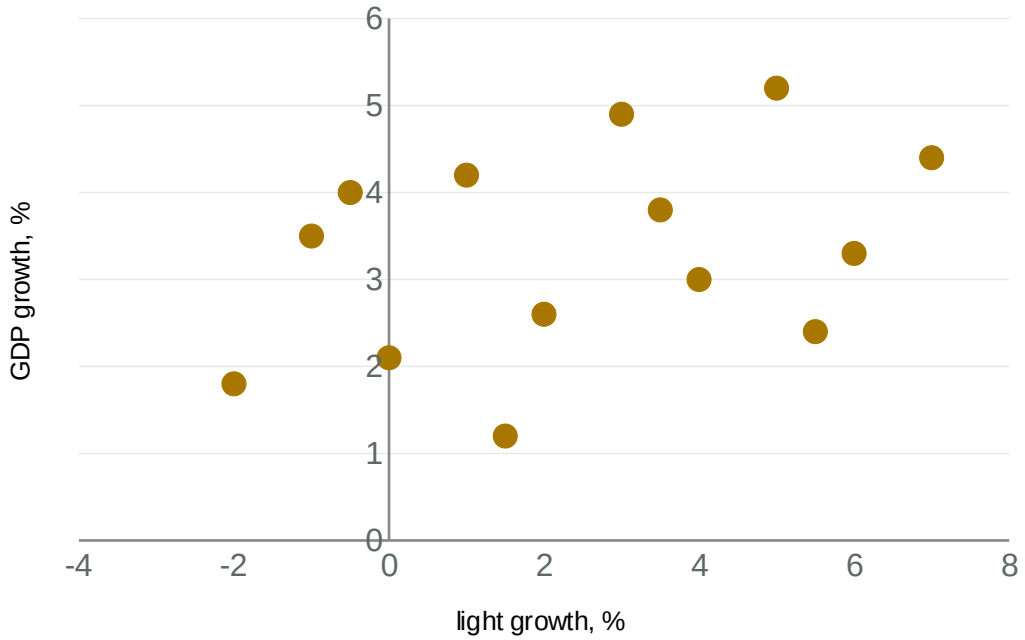
Adapt variable names to your files. For options B–D, swap the comparator and keep the levels-versus-changes logic.

Strong in levels, weaker in changes — the usual pattern

Levels: log GDP vs log Sum of Lights



Changes: GDP growth vs light growth (%)



Size drives both axes: big regions are bright and productive. A high correlation here proves little on its own.

The real test for monitoring: does light growth track output growth within the same region over time?

Illustrative synthetic data showing a typical pattern — not results for any country.

Report where the relationship holds — in one sentence

REPORTING CARD — ONE PER TEAM

Application & country	*e.g. Output proxy · regions of country X · 2013–2024*
Official comparator	*Source, unit, years, number of units*
Strength in levels	*Correlation / slope / R^2*
Strength in changes	*Correlation / slope / R^2 of growth rates*
Holds where …	*Regions, periods or sectors where the fit is good*
Breaks where …	*…and which link of the inferential chain explains it*
Decision	*Experimental statistic · diagnostic tool · not fit for purpose*

The one-sentence report

"Night-time lights track [official indicator] well in [where / when], because [lighting behaviour holds]; they break down in [where / when], because [link in the chain fails]. We classify the indicator as [decision]."

Two minutes per team. The cards are collected to seed follow-up project briefs.

SECTION 5

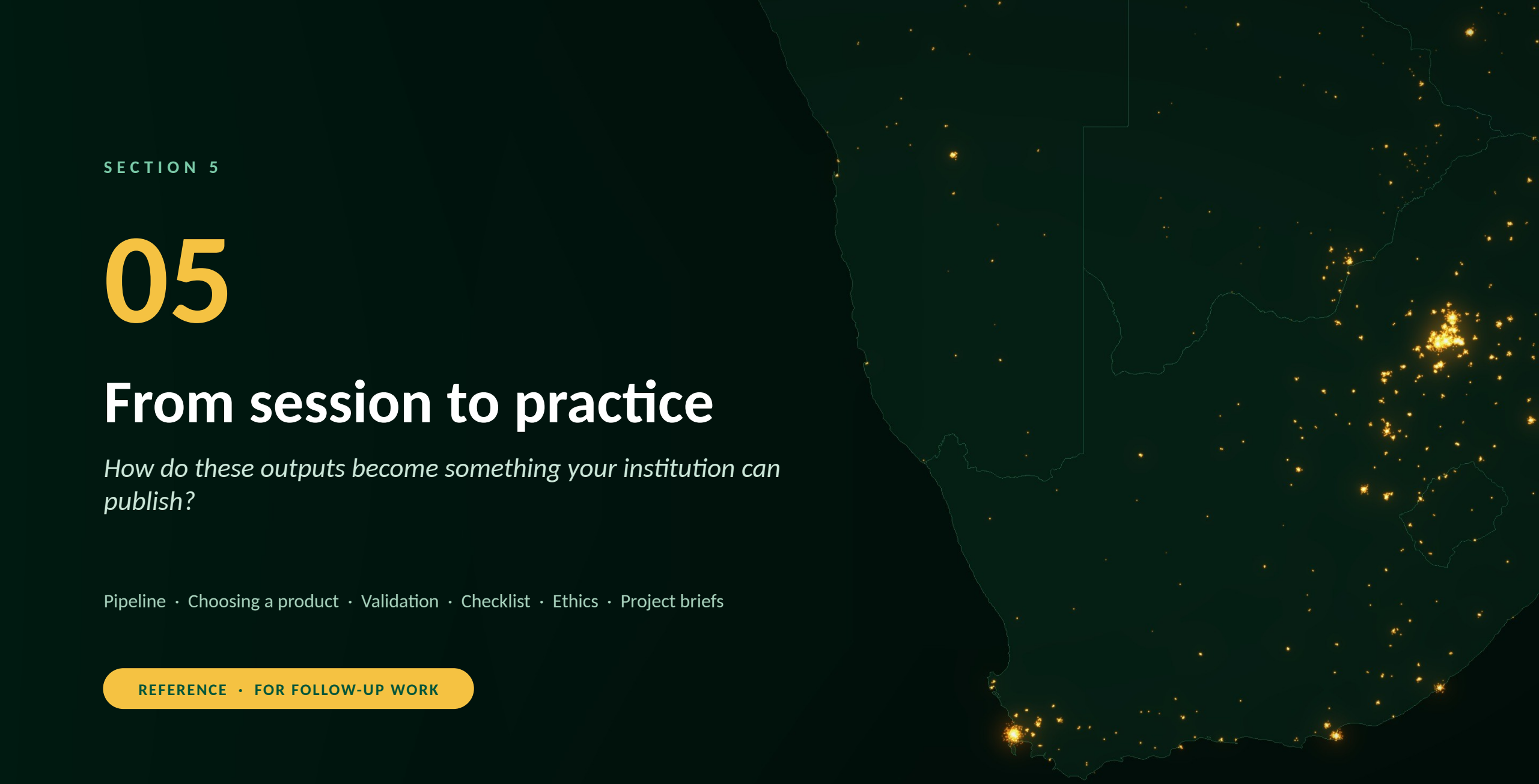
05

From session to practice

How do these outputs become something your institution can publish?

Pipeline · Choosing a product · Validation · Checklist · Ethics · Project briefs

REFERENCE · FOR FOLLOW-UP WORK



Southern Africa · Illustrative rendering: GeoNames settlement populations, Natural Earth borders

REFERENCE

Five stages, one reproducible pipeline

Collect

- NASA Earthdata / LAADS DAAC
- BlackMarblePy · Earth Engine
- Bands & quality flags

↓

National subset

Explore

- Histograms & maps
- Cloud & quality masks
- Month-to-month variation

↓

Artefact log

Analyse

- Zonal stats by admin level
- Annual & monthly series
- Change detection

↓

Tables & maps

Validate

- Official subnational data
- Census, surveys, utilities
- Fitness-for-use decision

↓

Evidence + decision

Publish

- Notebooks & maps
- Limitations statement
- README: method, sources, licence

↓

Public repository

Example open-source stack: BlackMarblePy / rasterio / xarray (read), geopandas (boundaries), exactextract or rasterstats (zonal statistics), statsmodels (validation), matplotlib (maps).

Which cadence for which question?

Annual

VNP46A4 · EOG VNL annual

STRENGTH

Cleanest series; noise and ephemeral lights largely filtered.

WATCH OUT FOR

A change cannot be dated within the year.

TYPICAL USE

Subnational panels, long-run trends, urban expansion, poverty mapping, GDP downscaling.

Monthly

VNP46A3 · EOG VNL monthly

STRENGTH

Dates a change to within a season; feeds quarterly nowcasts.

WATCH OUT FOR

Rainy-season months may rest on very few valid nights.

TYPICAL USE

Nowcasting, programme evaluation, seasonal activity, conflict & recovery timelines.

Daily

VNP46A2

STRENGTH

Every night, corrected for moon, atmosphere and terrain.

WATCH OUT FOR

Large volumes; a single night is noisy — aggregate first.

TYPICAL USE

Blackouts, disaster response, reliability of electricity supply, event studies.

Rule of thumb: start annual. The finer the cadence, the more corrections you must trust and the more noise you must handle.

REFERENCE

Publish, or keep as a diagnostic?

Q1 — **Is the relationship strong and stable across units and years?**

Set your bar before you look.

Q2 — **Are artefacts identified and treated?**

Flares, blooming, seasonality, cloud, noise.

Q3 — **Is the benchmark independent?**

WorldPop uses NTL as a covariate — circular.

All three yes → Experimental statistic — publish with limitations

Partly → Diagnostic tool — internal use only

No → Not fit for purpose — document why

Limitations statement — template

Part of the deliverable, committed to GitHub

Data	Product, collection version, period, cadence
Choices	Boundaries, radiance threshold, masks applied
Artefacts	Which of the six were found, where, and how treated
Validation	Benchmark, year, number of units, strength in levels & changes
Decision	Experimental, diagnostic or not fit — and why
Must not be used for	e.g. household income, small-settlement counts

Suggested guide — adapt to your national quality assurance framework. Circularity: see Falchetta et al. (2019), Scientific Data.

A checklist for defensible analysis

1 Name product and version

VNP46A3 collection 2, VNL V2.x — they are not interchangeable.

2 Never splice sensors naively

Any DMSP–VIIRS series needs documented harmonisation.

3 Report and test the threshold

Show the finding survives a reasonable range.

4 Mask what you are not studying

Background noise, gas flares, fishing lights, fires.

5 Normalise deliberately

Per capita, per km², or neither — state and defend it.

6 Validate against the ground

Census, surveys, utility records — independent sources.

7 Prefer changes to levels

Differences cancel much of the calibration error.

8 Publish code and limitations

README with method, sources, licence and caveats.

THE GOLDEN RULE

State, in one sentence, what would have to be true about lighting behaviour for your conclusion to hold.

If you cannot write that sentence, you do not yet have a result.

Share it: a documented, reusable repository is how your institution contributes to a shared community of practice.

REFERENCE

Ethics, privacy and governance for night-lights statistics

Privacy

At ≈ 500 m, lights rarely identify individuals. Risk rises when they are combined with building footprints or high-resolution imagery for very small settlements.

Conflict sensitivity

Maps of darkening can reveal displacement or vulnerable populations. Consider aggregation and release timing before publishing.

Licensing & sovereignty

NASA and EOG products are open; commercial night imagery often is not. Check licences before redistributing derived products.

Labelling

Publish as experimental statistics with a limitations statement; never present a proxy as the official measure.

Coordination

Agree with the national statistical system who produces, validates and publishes — avoid parallel numbers.

AI & models

If machine-learning models use lights, document features and training data, and apply your AI governance rules.

Builds on good practice in data access, governance, AI, ethics and privacy, and on the UN Fundamental Principles of Official Statistics.

Six night-lights project briefs for your institution

Subnational activity index

Monthly SOL by region, masked and validated against regional accounts.

Output: Experimental regional indicator

Nowcasting feature

NTL growth added to a GDP nowcasting model; test the gain out-of-sample.

Output: Nowcast with and without lights

Electricity reliability monitor

Share of lit nights per district from daily data, next to official access rates.

Output: Reliability dashboard

Urban expansion monitor

Lit area and radiance growth combined with built-up grids.

Output: Sprawl vs densification map

Rapid outage assessment

Event-study template for cyclones, floods or grid failures.

Output: Reusable notebook + brief

Credibility cross-check

Compare regional GDP growth with light growth; investigate the largest gaps.

Output: Residual map + memo

Each brief states its question, data sources and intended output · small enough to prototype in a few days.

Five questions to check your understanding

1	**Why can a city look dimmer to VIIRS after switching to LED street lights?**	Its blue peak (≈ 450 nm) lies outside the Day/Night Band.
2	**A district's lit area fell 25% when you changed the threshold from 0.5 to 1.0. What changed in the world?**	Nothing — it is a researcher choice. Report and test it.
3	**Your NTL indicator correlates 0.9 with WorldPop. Is it validated?**	No — WorldPop uses NTL as a covariate. The benchmark is not independent.
4	**Correlation is high in levels but weak in growth rates. What does that mean?**	Size drives levels; the indicator is not yet a reliable monitoring tool.
5	**A coastal province goes dark for a month in the rainy season. First check?**	The number-of-observations layer: cloud, not collapse.

Tip for presenters: hide the right-hand column with a shape or reveal answers one by one.

IN CLOSING

Five things to take into the labs

1. Night lights are a proxy, never the quantity of interest — name the assumption that links them to your question.
2. The instrument defines the science: DMSP saturates, VIIRS calibrates, and neither sees blue light.
3. The data is most valuable where official statistics are thinnest: subnational, high-frequency, in crises and informal economies.
4. Darkness is not emptiness — off-grid light, cloud and load-shedding all look alike from orbit.
5. Validate against independent sources, in levels and in changes, and publish your limitations with your results.



The darkness tells us a great deal. It does not tell us everything — and it never speaks unprompted.

Mauritius · Illustrative rendering: GeoNames settlement populations, Natural Earth borders

The vocabulary you will meet in this module

Radiance

Energy leaving a surface per unit area and solid angle; for NTL in $nW \cdot cm^{-2} \cdot sr^{-1}$.

Digital number (DN)

Uncalibrated sensor count with no physical unit; DMSP reports 0–63.

Composite

Grid assembled from many cloud-free nights — not a photograph of one night.

Stable lights

Product with fires, aurora and other unstable sources removed.

Sum of Lights (SOL)

Total radiance summed over all pixels inside a boundary.

Blooming / overglow

Apparent spread of light beyond its source (scattering, geolocation error).

BRDF correction

Adjustment for how reflected (moon)light varies with viewing and lighting angles.

Zonal statistics

Summary of raster values (sum, mean, count) within each polygon.

Nadir

Point directly beneath the satellite, where the pixel footprint is smallest.

Elasticity

% change in output associated with a 1% change in light; not a constant.

Selected references

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Data, tools and tutorials to keep

NASA Black Marble

LAADS DAAC · Earthdata — VNP46 / VJ146 products

Earth Observation Group

VNL annual & monthly composites; VIIRS Nightfire

Google Earth Engine

Hosted, analysis-ready EOG and Black Marble collections

BlackMarblePy / blackmarbler

World Bank packages for Python and R

HREA

Settlement-level electricity access & reliability

Light Every Night

Every nightly DMSP & VIIRS observation, open on AWS

GHSL · WSF · Open Buildings

Built-up grids to pair with lights

World Bank Open Nighttime Lights

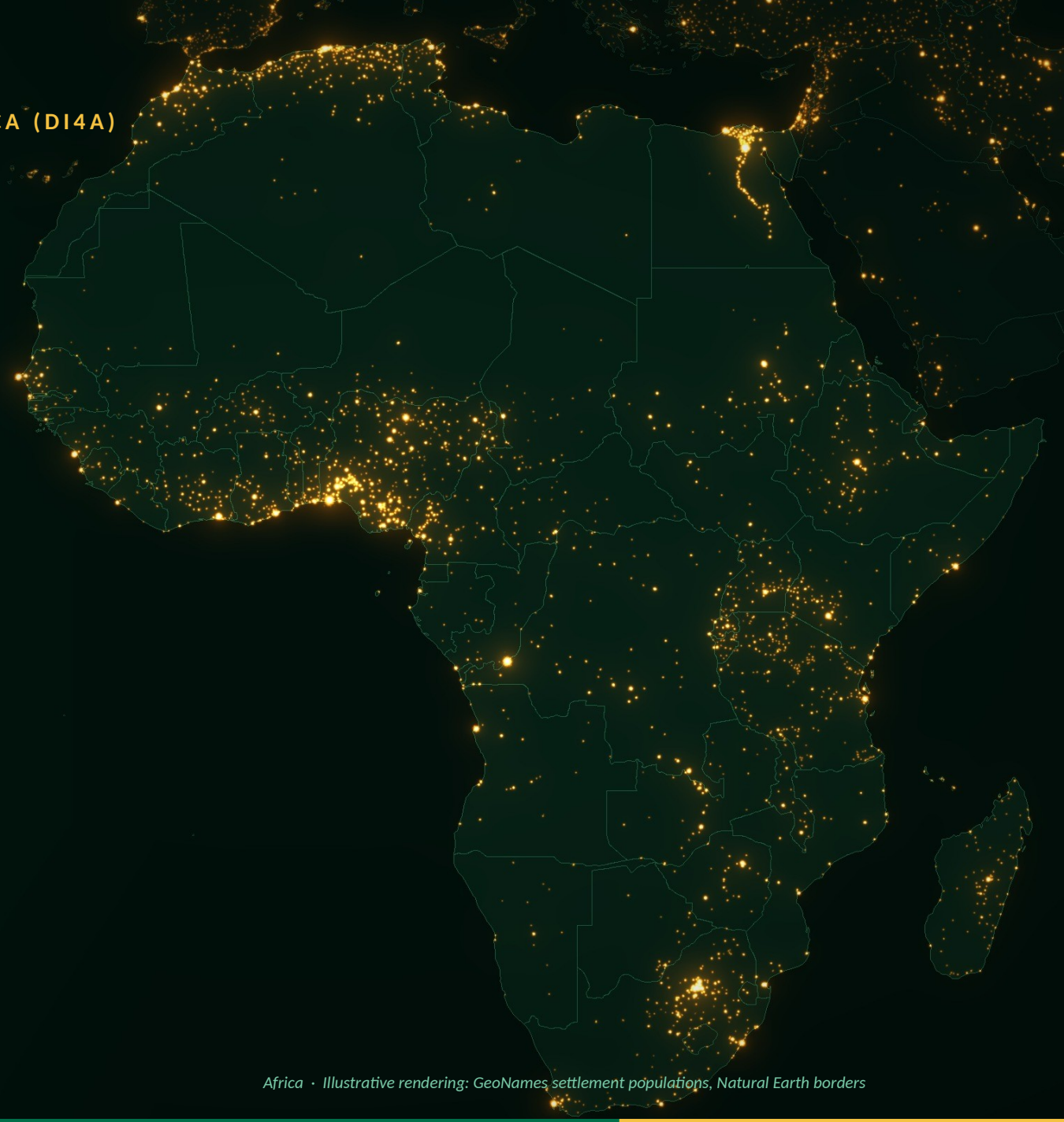
Tutorials, notebooks — and most real images in this deck (CC BY 4.0)

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Thank you

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Africa · Illustrative rendering: GeoNames settlement populations, Natural Earth borders